# PRACTICAL NO. 1

## Aim : To import the student dataset into Power BI, understand the difference between Load and Transform Data, create a Clustered Column Chart, and apply a slicer for dynamic filtering.

### Step 1: Start Power BI Desktop

- When Power BI Desktop is opened, the **Home screen** appears.

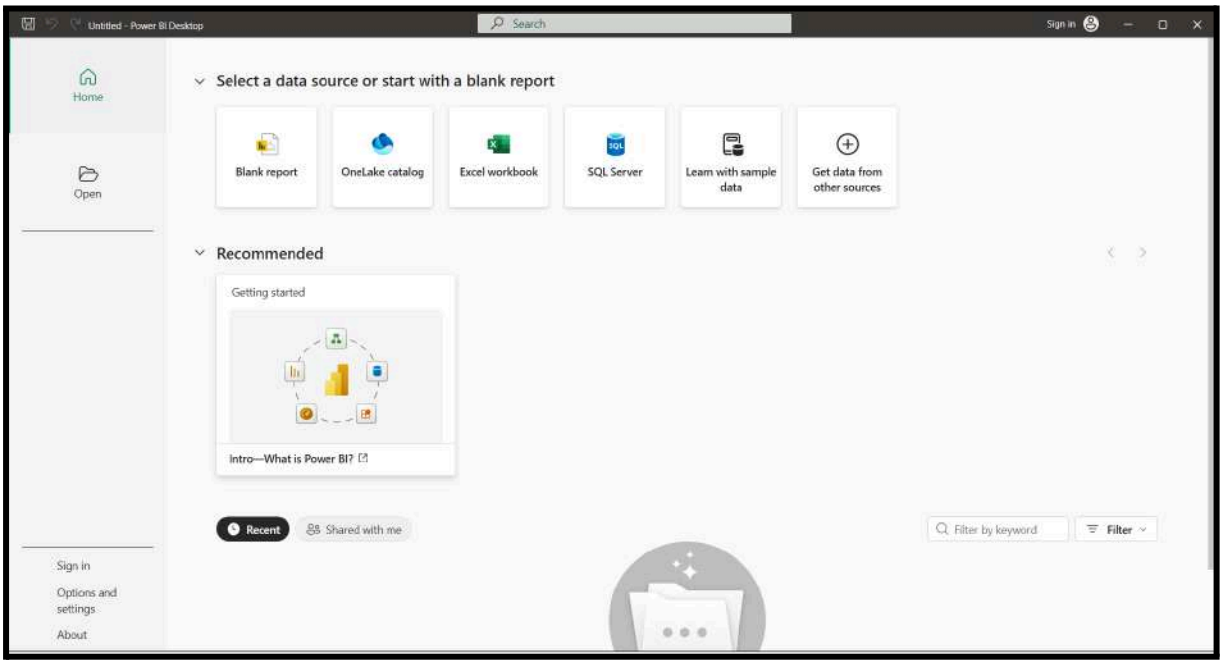



Fig. 1: Power BI Desktop Home Window

### Step 2: Import Data

- Click Excel Workbook from the Home screen.
- Select the dataset file → Click **Open**.
- A **Navigator Window** appears showing the sheets inside the workbook.

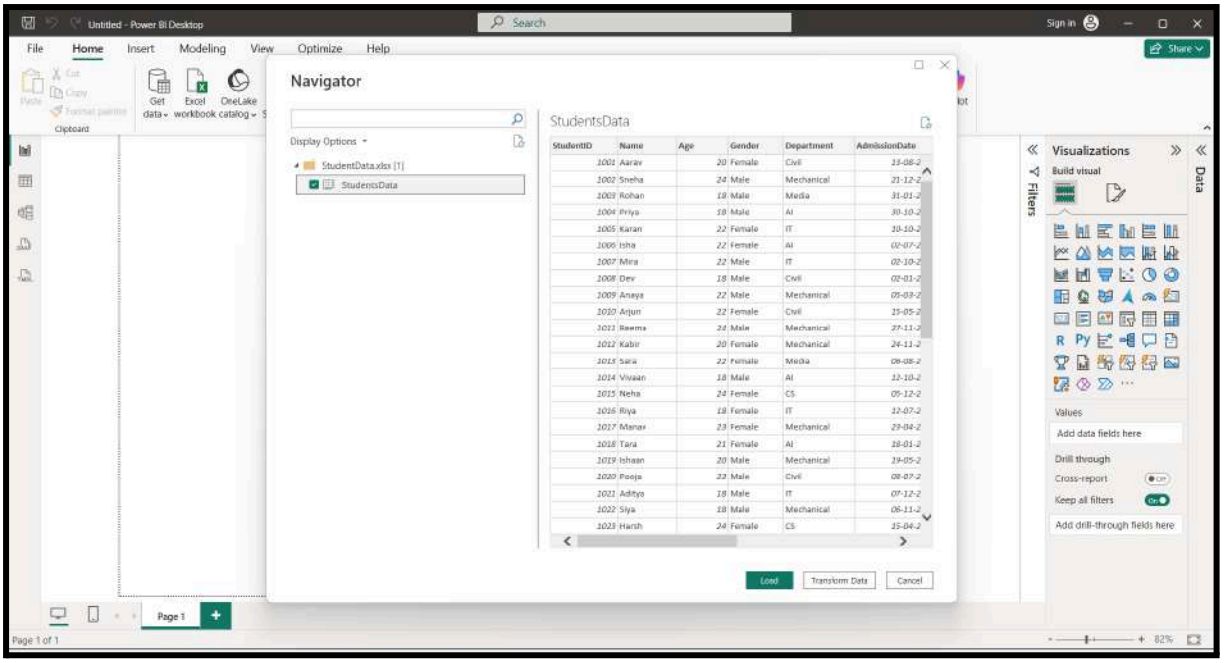



Fig. 2: Power BI Desktop Navigator Window

**Load vs Transform Data (When to Use What)**

- **Use Load** when:
  - Data is already clean
  - No missing values
  - No column name changes needed
  - Data types (number/text/date) are correct
  - Direct import is sufficient
- **Use Transform Data** when:
  - You need to clean the dataset
  - Remove blank rows
  - Replace values
  - Convert data types
  - Split or merge columns
  - Filter specific rows
  - Add calculated columns

Choosing **Transform Data** opens the **Power Query Editor**.

**Step 3: Load the Data**

- After selecting a sheet:
  - Click **Load** (if data is clean)
  - OR click **Transform Data** → Apply cleaning → Click **Close & App**
- The dataset fields now appear in the **Fields Pane** on the right side.

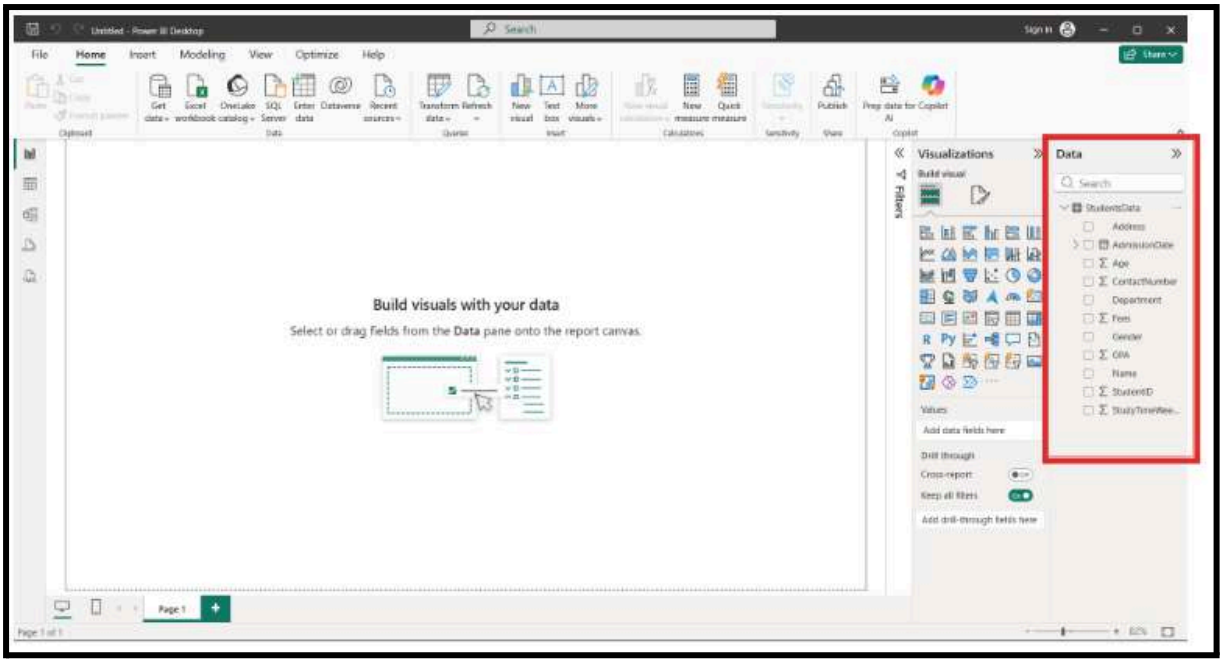



Fig. 3: Loaded Fields in Power BI

**Step 4: Create a Bar Graph**

- Go to the **Visualizations Pane**.
- Click **Clustered Column Chart** icon

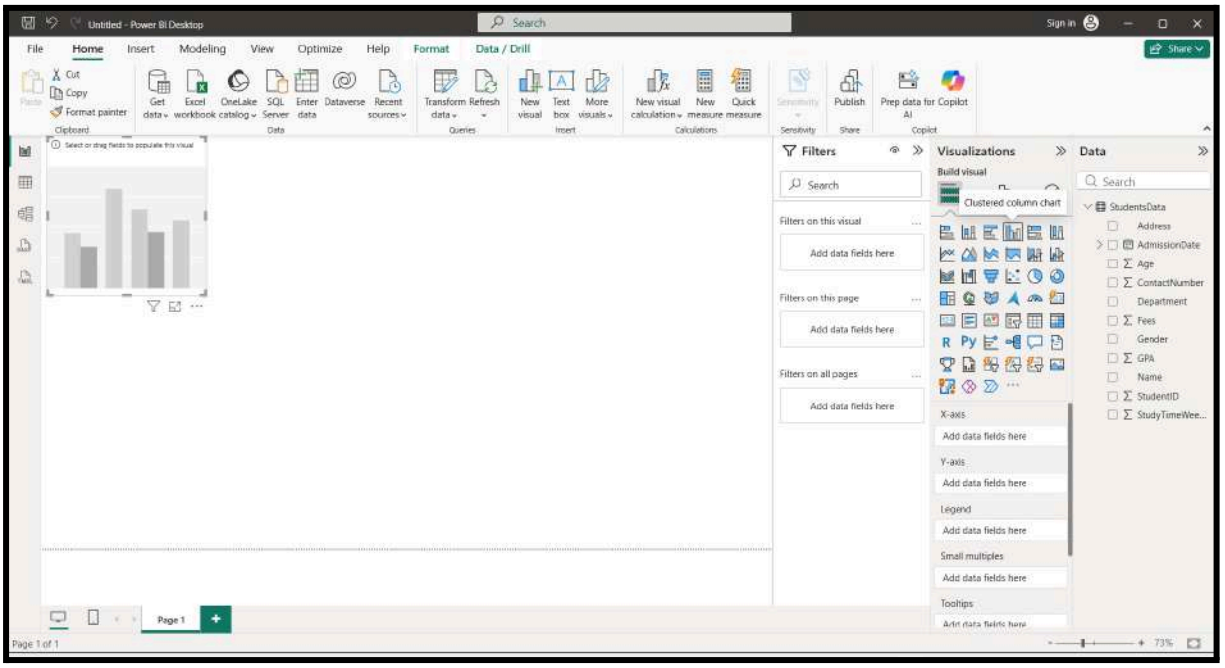



Fig. 4: Column Chart Icon

- Set the fields:
- **Y-Axis** → *Count of Name*
- ***X-Axis***→ *Department*

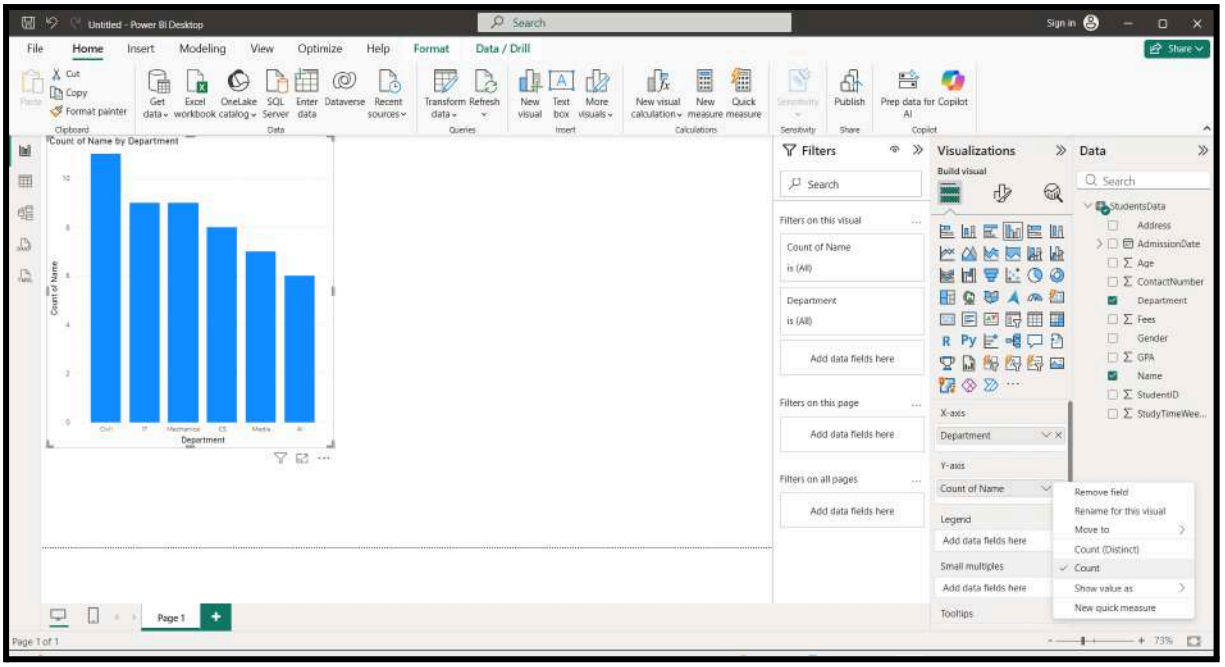



Fig. 5: Axis and Value Fields Assignment

### Step 5: Add a Slicer

- Click the **Slicer** icon from Visualizations.

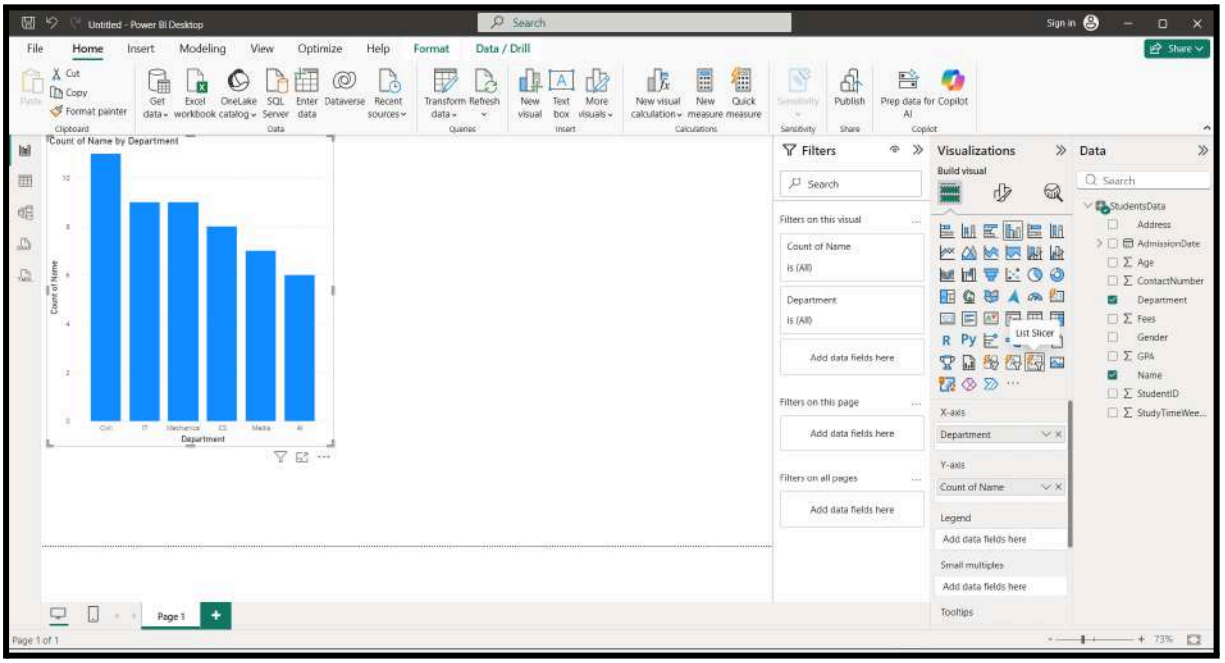



Fig. 6: List Slicer Icon

- A blank slicer appears.
- Drag any field into the slicer, e.g.:
    - **Gender** (Male/Female filtering)
    - **Address** (Location-wise filtering)

The slicer will now filter the Clustered Column Chart dynamically.

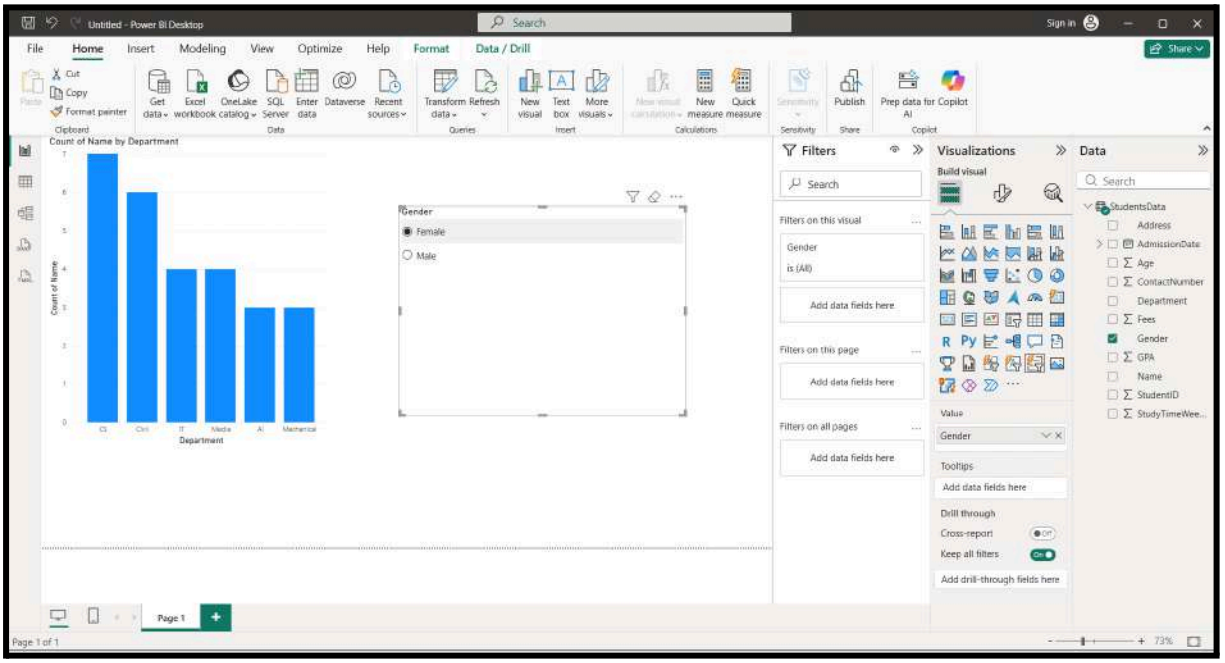



Fig. 7: Adding a Slicer

**Step 6: Format the Clustered Column Chart**

Click on the chart → Open Format (paint roller icon).

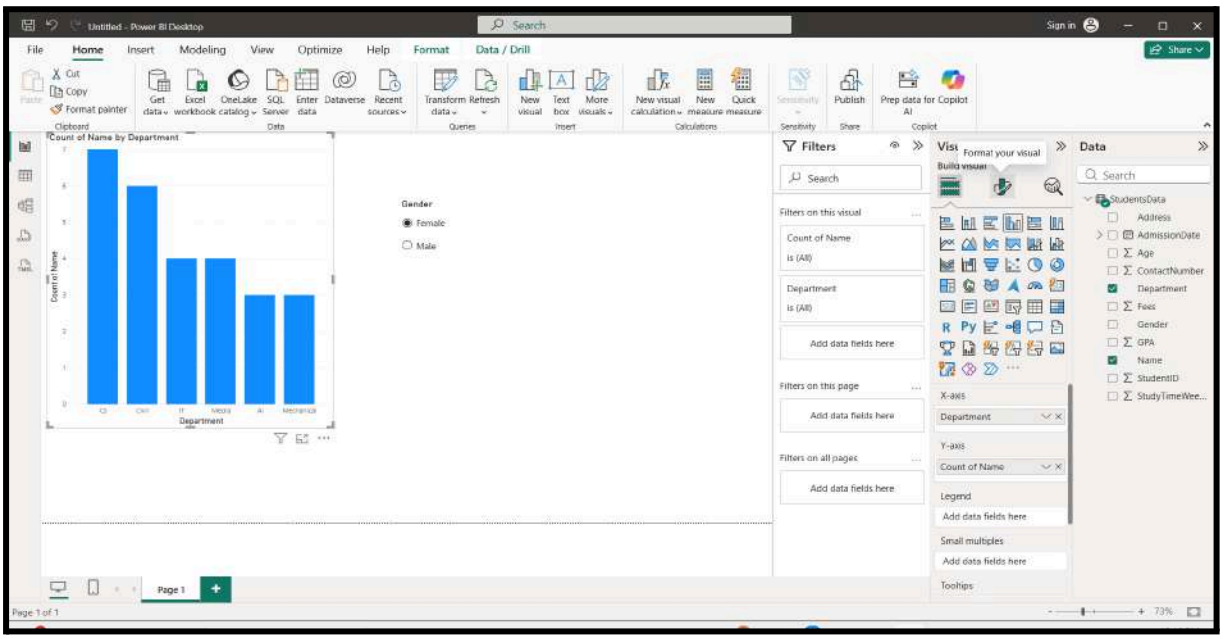



Fig. 8: Format Option

6.1 Change Data Colors (Bar Colors):

- Format → **Columns → Colors → Choose color**

6.2 Add Border & Background:

- Format → **Border → On → Choose border color**

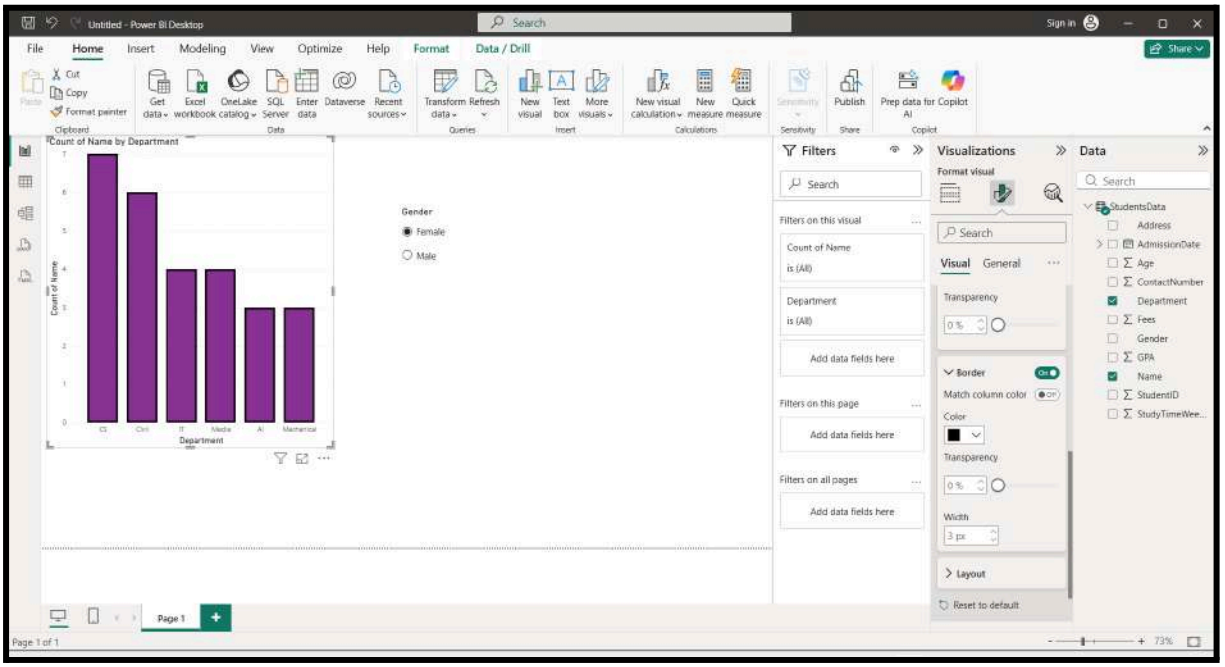



Fig. 9: Formatted Chart

**Step 7: Save the Report**

- Go to **File → Save As**
- Save the file

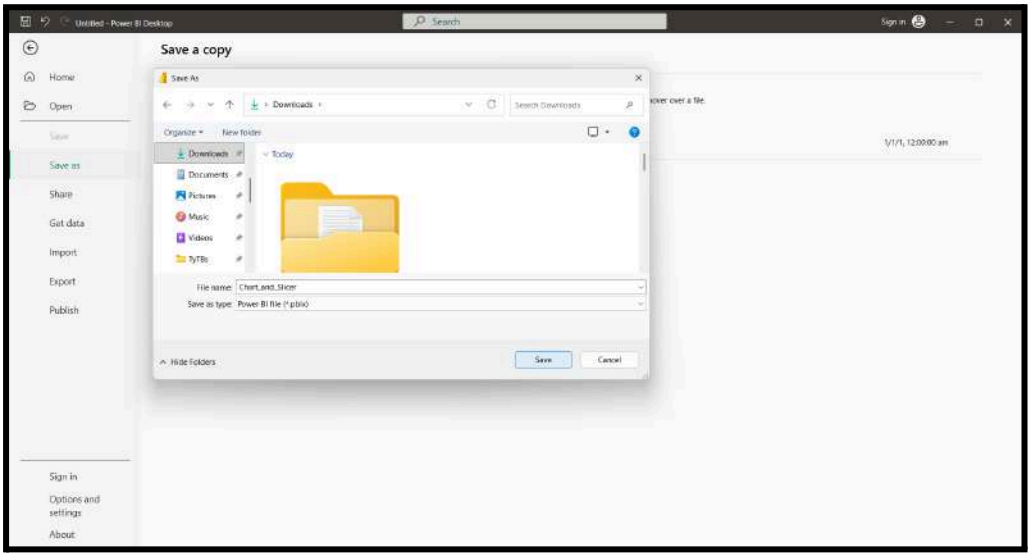
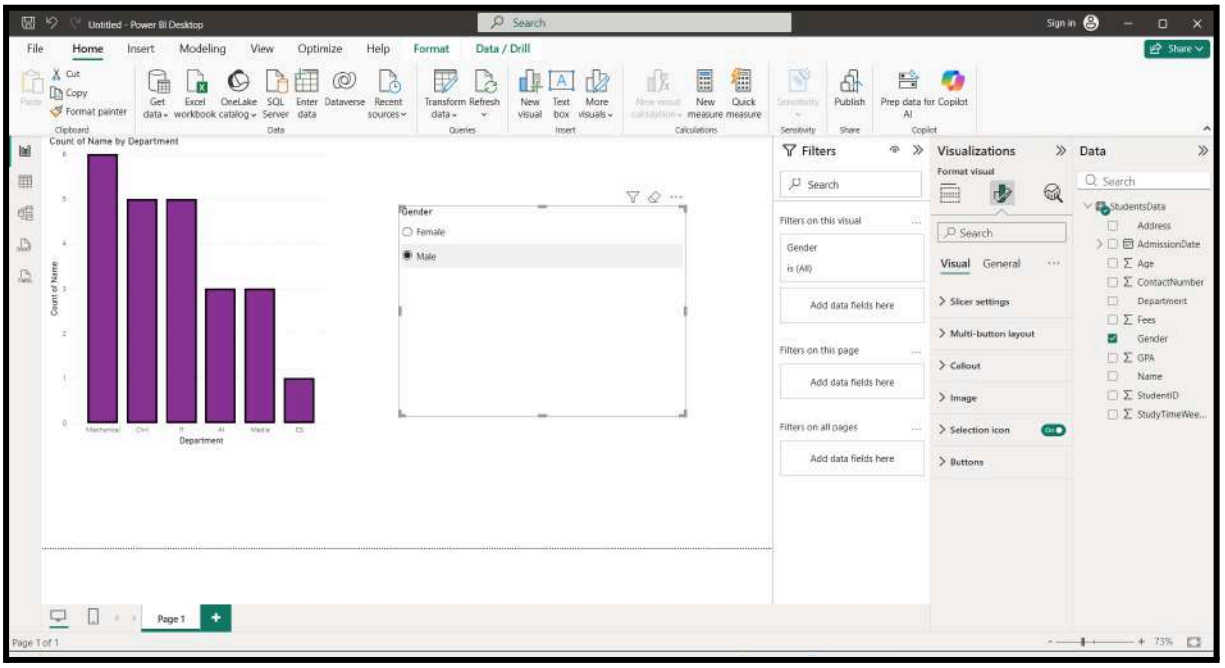



Fig. 10: Saving the File

# Output:

A fully formatted **Clustered Column Chart** showing department-wise student count, controlled by a **dynamic slicer**.



# PRACTICAL NO. 2

## Aim : Creating a Pie Chart with Grouping and Formatting in Power BI

### Step 1: Start Power BI Desktop

- When Power BI Desktop is opened, the **Home screen** appears.

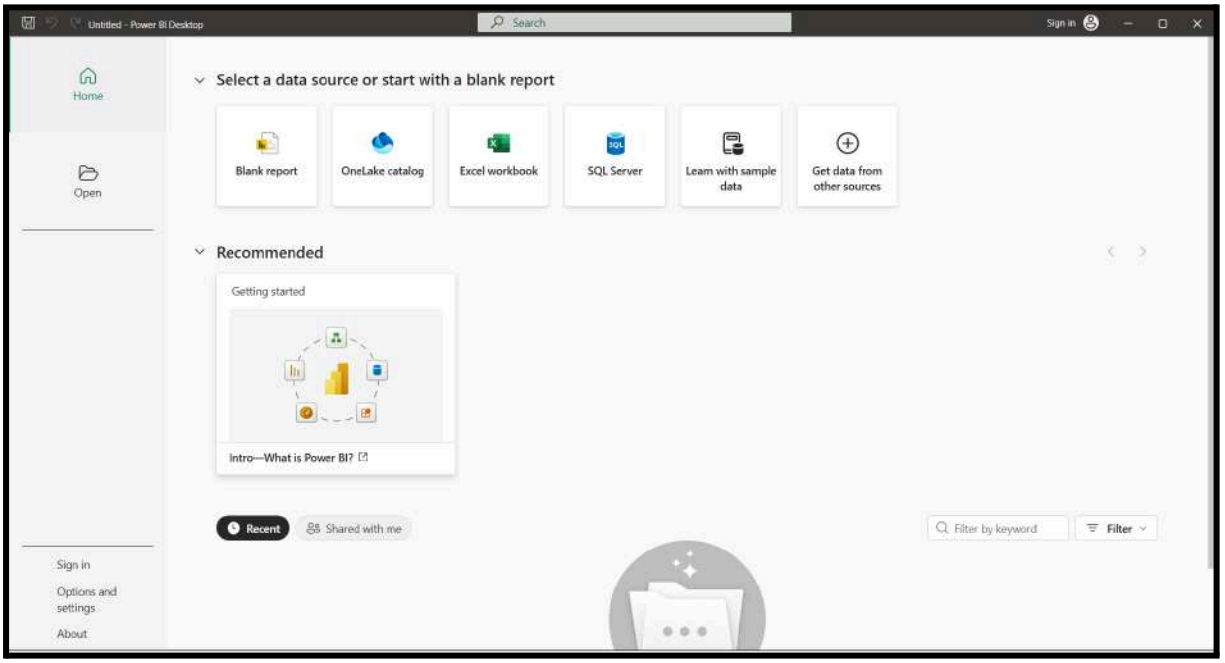



Fig. 1: Power BI Desktop Home Window

### Step 2: Import Data

- Click Excel Workbook from the Home screen.
- Select the dataset file → Click **Open**.
- A **Navigator Window** appears showing the sheets inside the workbook.

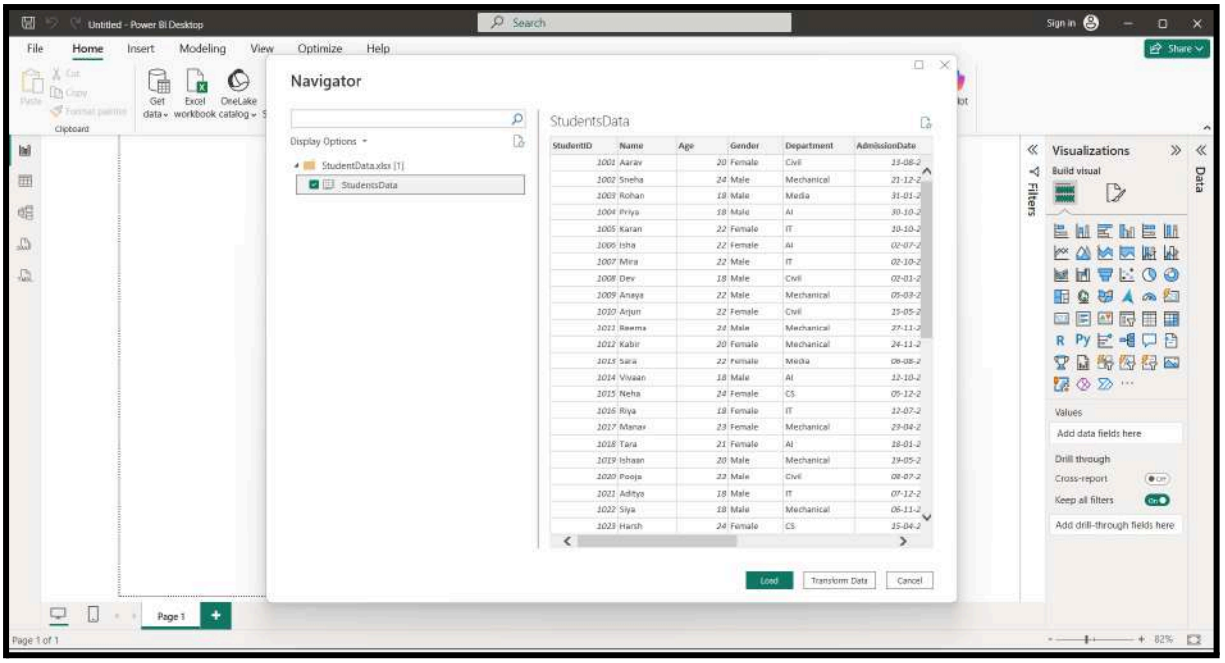



Fig. 2: Power BI Desktop Navigator Window

**Step 3: Load the Data**

- After selecting a sheet:
  - Click **Load** (if data is clean)
  - OR click **Transform Data** → Apply cleaning → Click **Close & App**
- The dataset fields now appear in the **Fields Pane** on the right side.

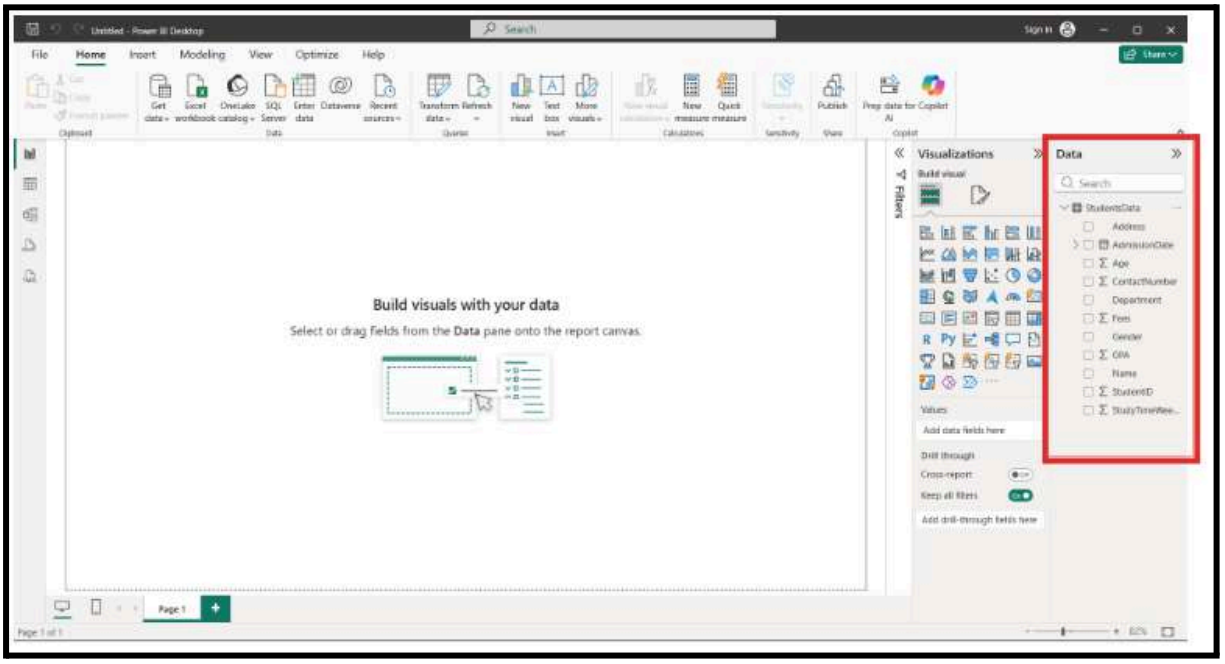


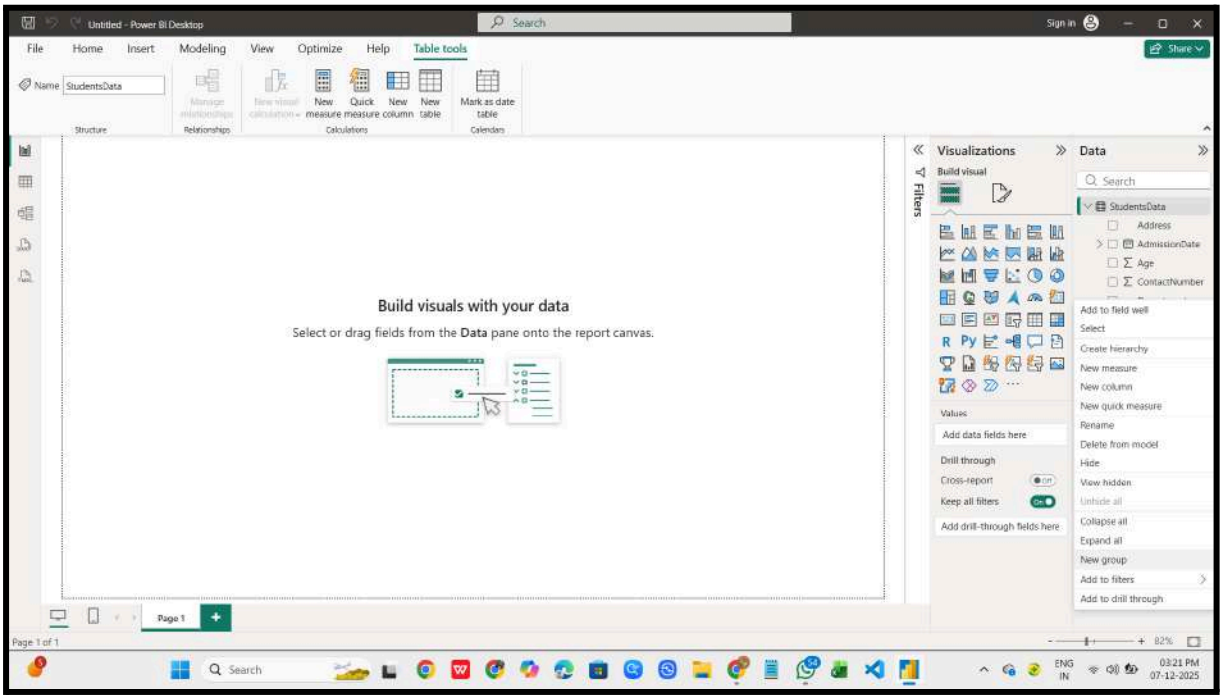

Fig. 3: Loaded Fields in Power BI

**Step 4: Group Data for Pie Chart**

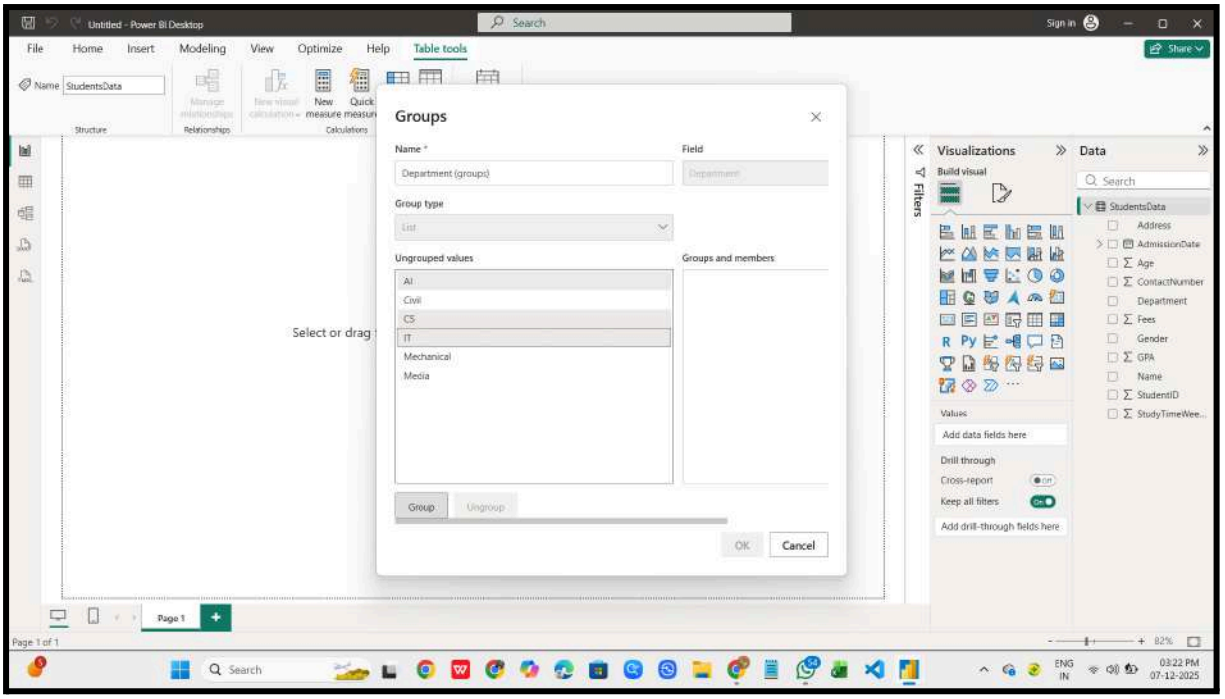
To Group Data:

1. Go to **Data View** (left Table icon)

2. Right-click **Department** column → Choose **New Group**




3. Add items into custom groups




4. Give the group a name

5. Click **OK**

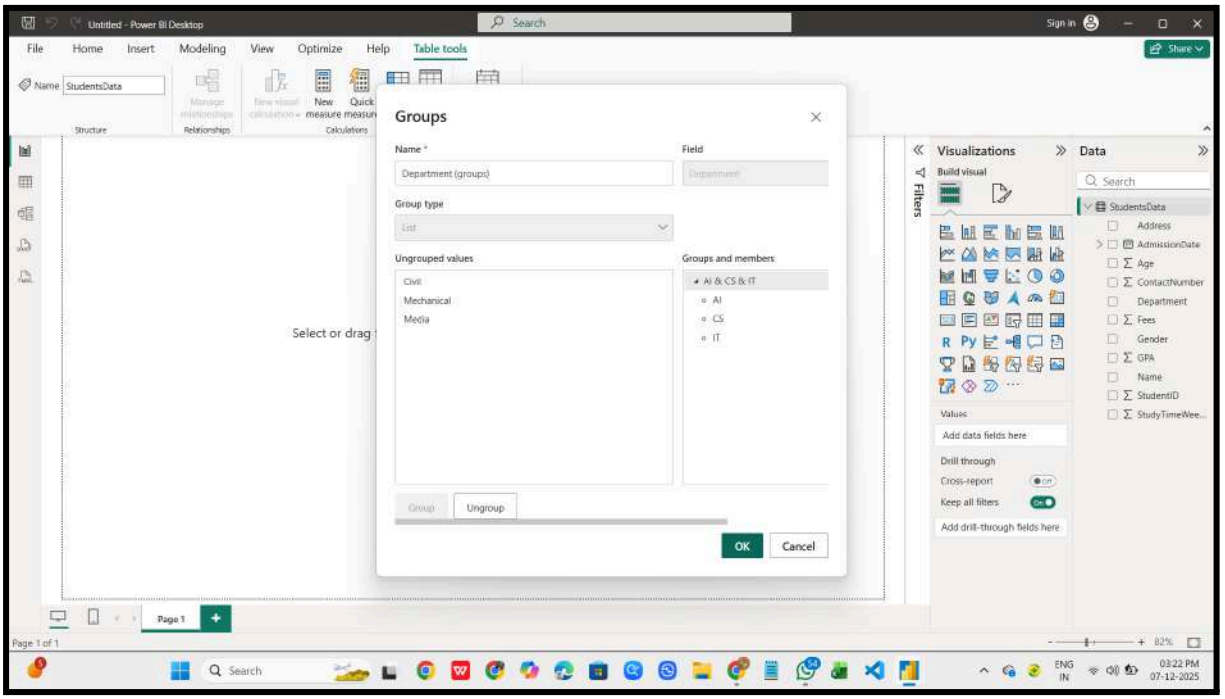



Fig. 4: Grouping Data

A new field *Department (groups)* will appear.

**Step 5: Insert Pie Chart**

1. Go to **Visualizations Pane**
2. Click the **Pie Chart** icon
3. A blank pie chart appears on the canvas

**Assign Fields:**

- Legend → **Gender**
- Values → **StudentID (Count)**

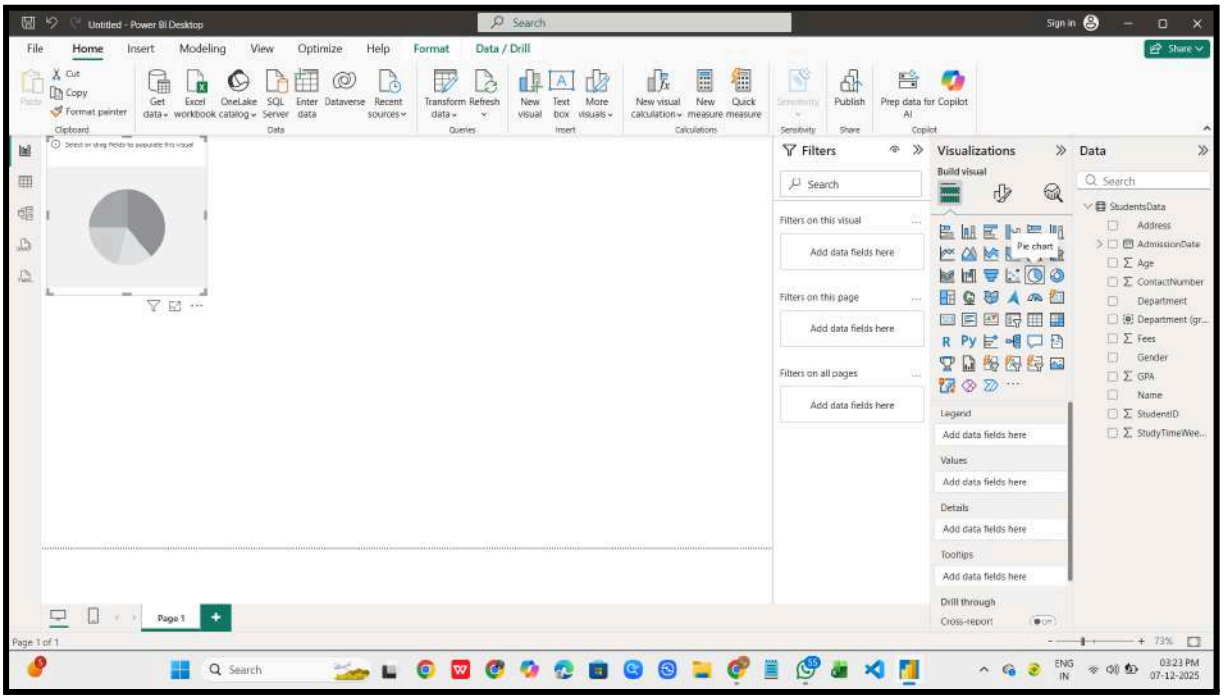



Fig. 5: Pie Chart Icon

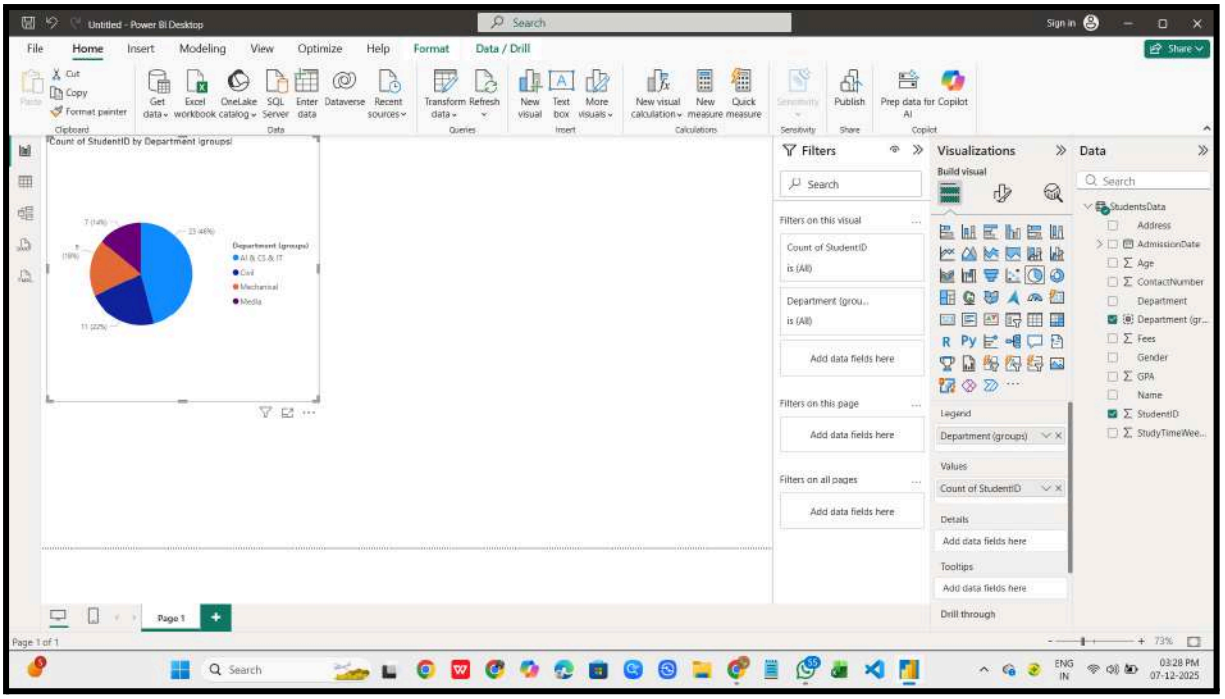



Fig. 6: Assigning Fields to Pie Chart

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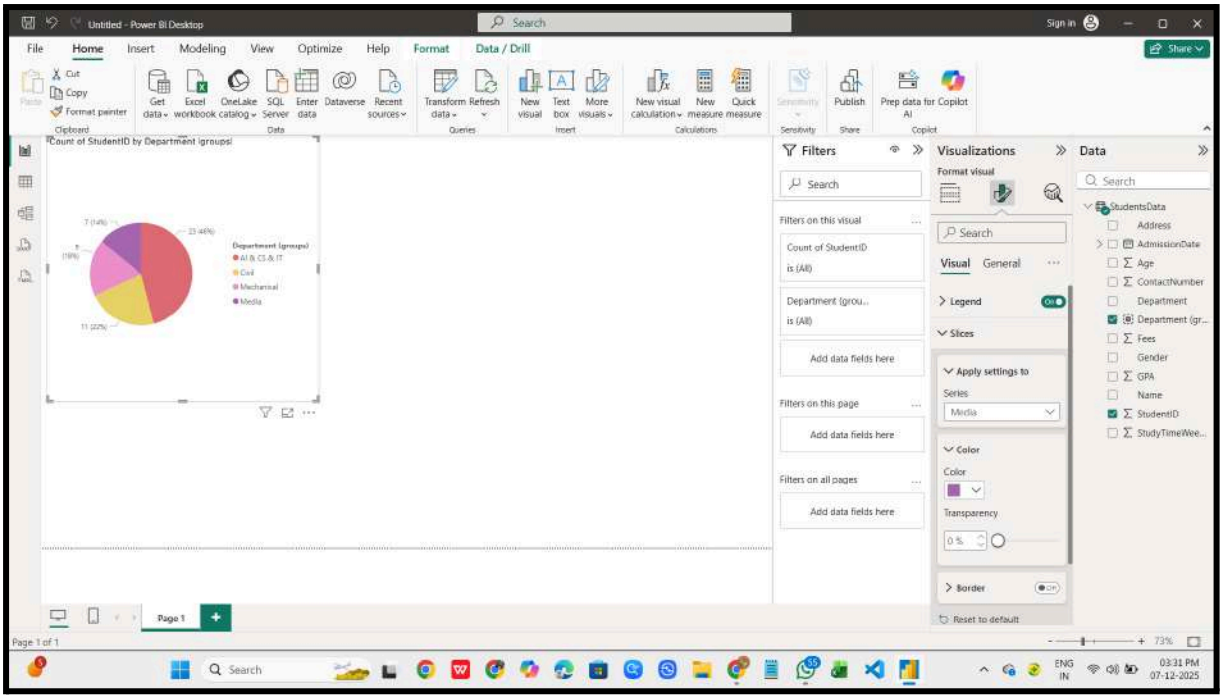
### Step 6: Format the Pie Chart

Click on the chart → Go to **Format (paint roller icon)**.

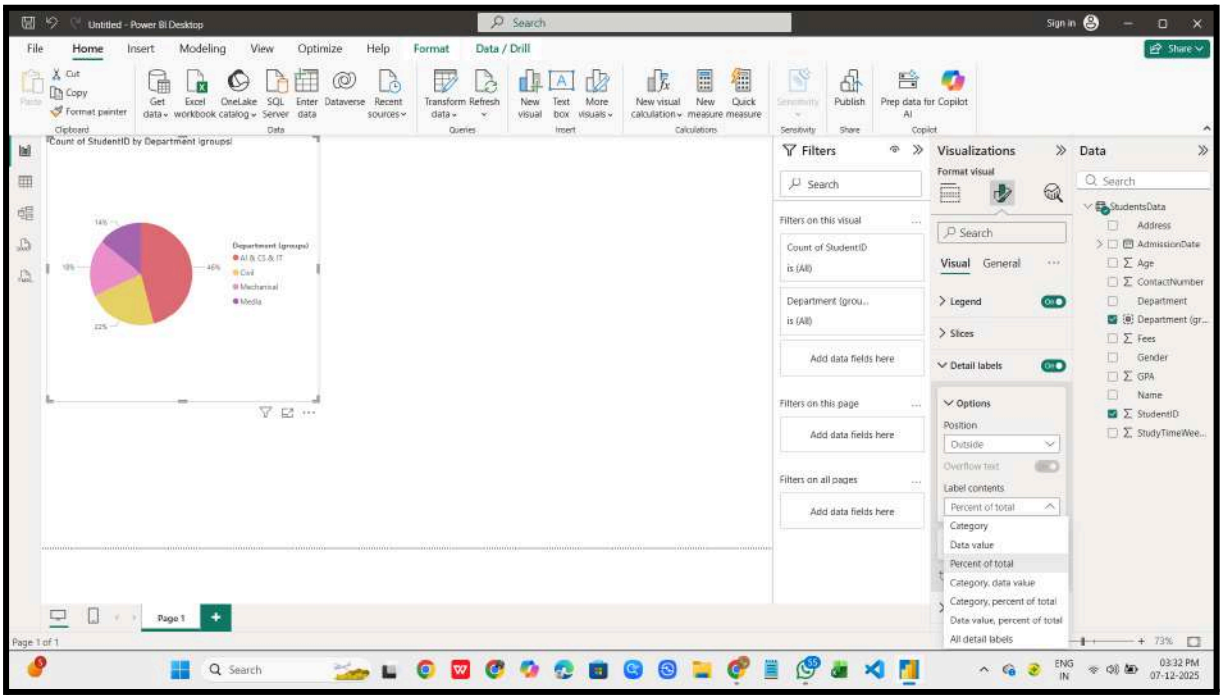
### 6.1 Change Slice Colors

- Format → **Slices → Series → Choose a value**
- Then go to **Colors → Choose Color**




## 6.2 Display Data Labels (Percentages)

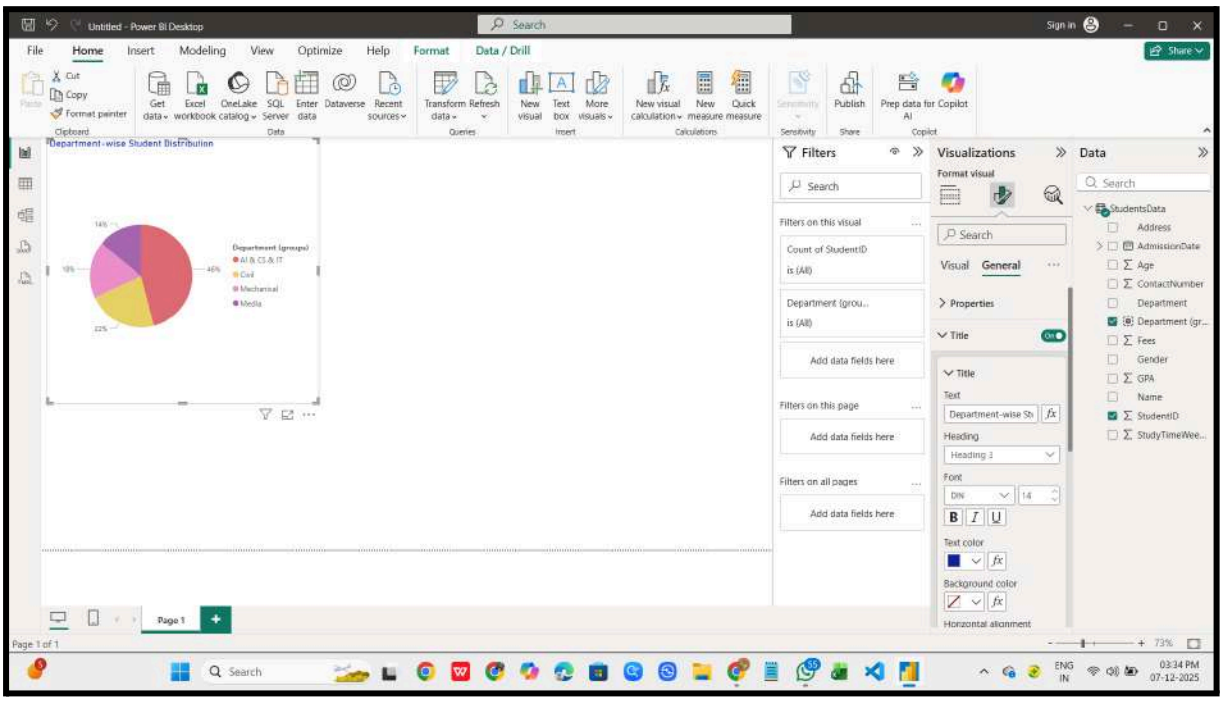
- Format → **Detail Labels → On → Label contents → choose "Percent of total"**




## 6.4 Format Title

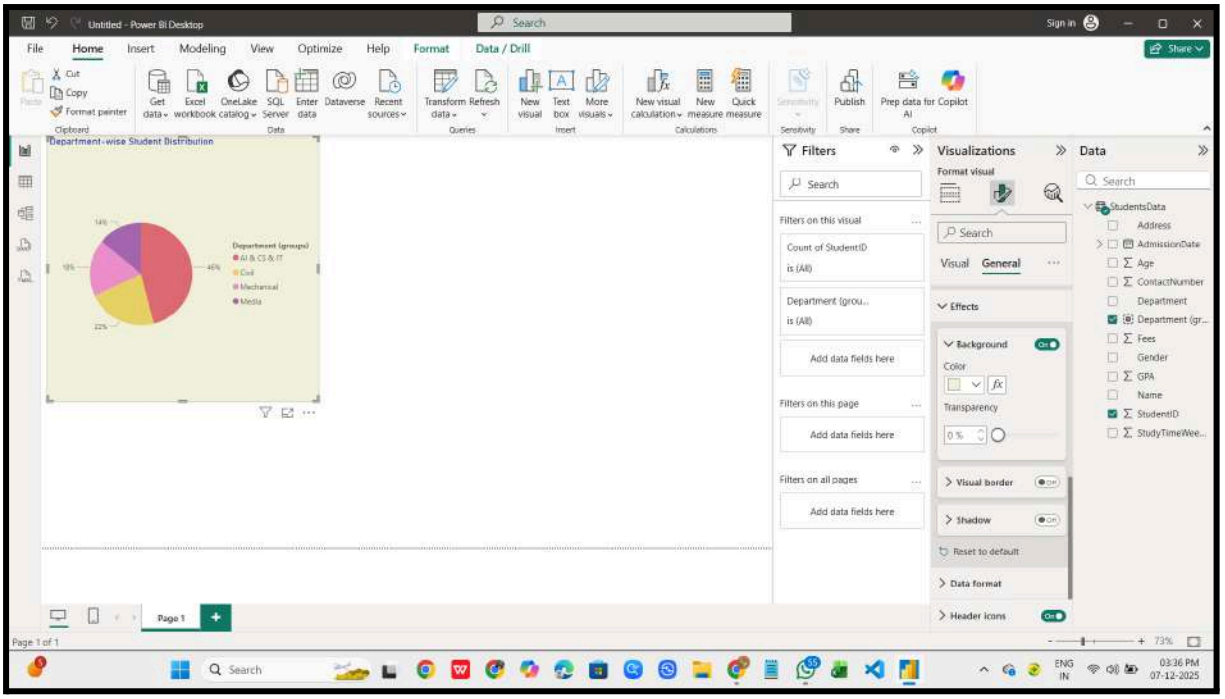
- Format → **Title → On**

- Title Text: **“Gender-wise Students Distribution”**
- Font Size: 18–20
- Font Color: Dark Blue



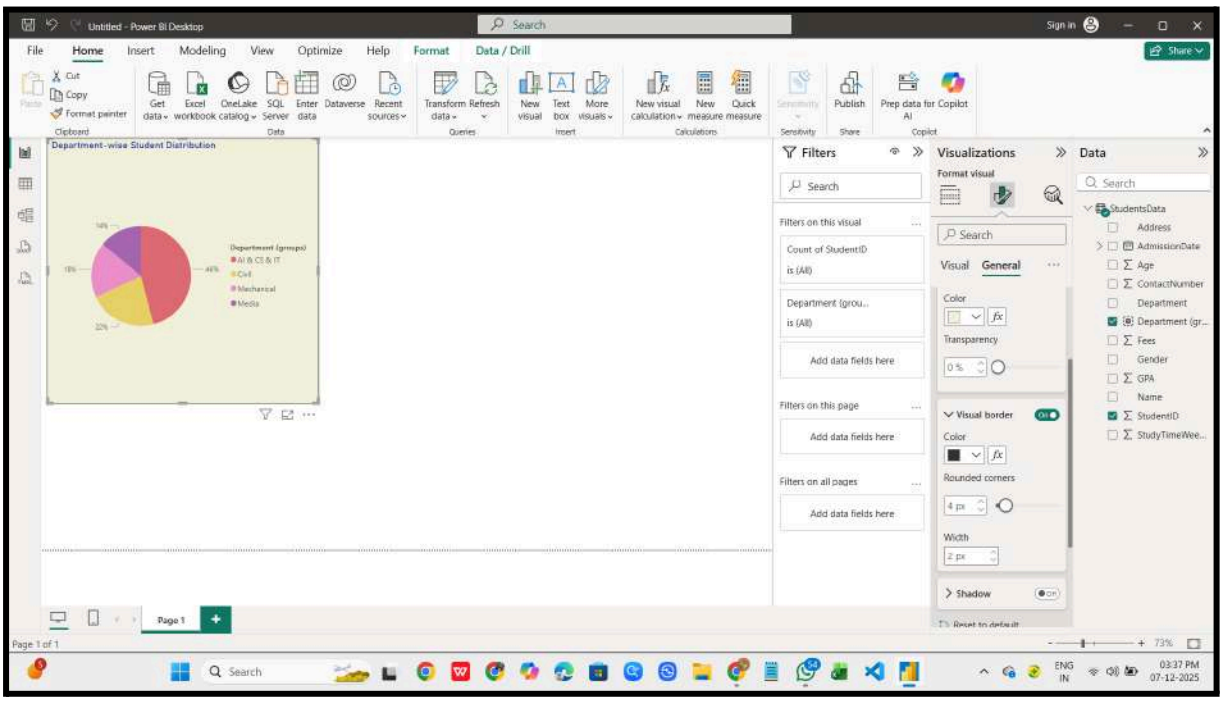

## 6.5 Background Formatting

- Format → **Background**
- Choose a soft light color
- Reduce transparency (20%-40%)




## 6.5 Add Border

- Format Visual → **General → Effects → Visual Border → On**
- Choose Any Color




## Step 7: Save the Report

- Go to **File → Save As**

- Save the file

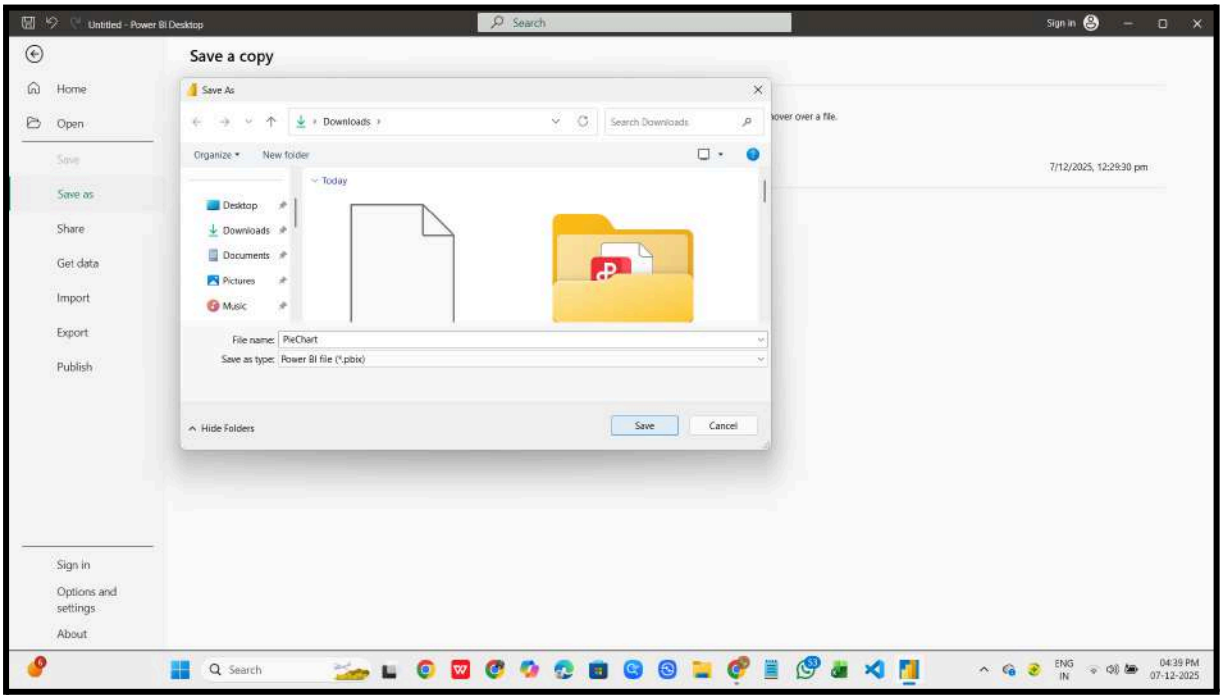
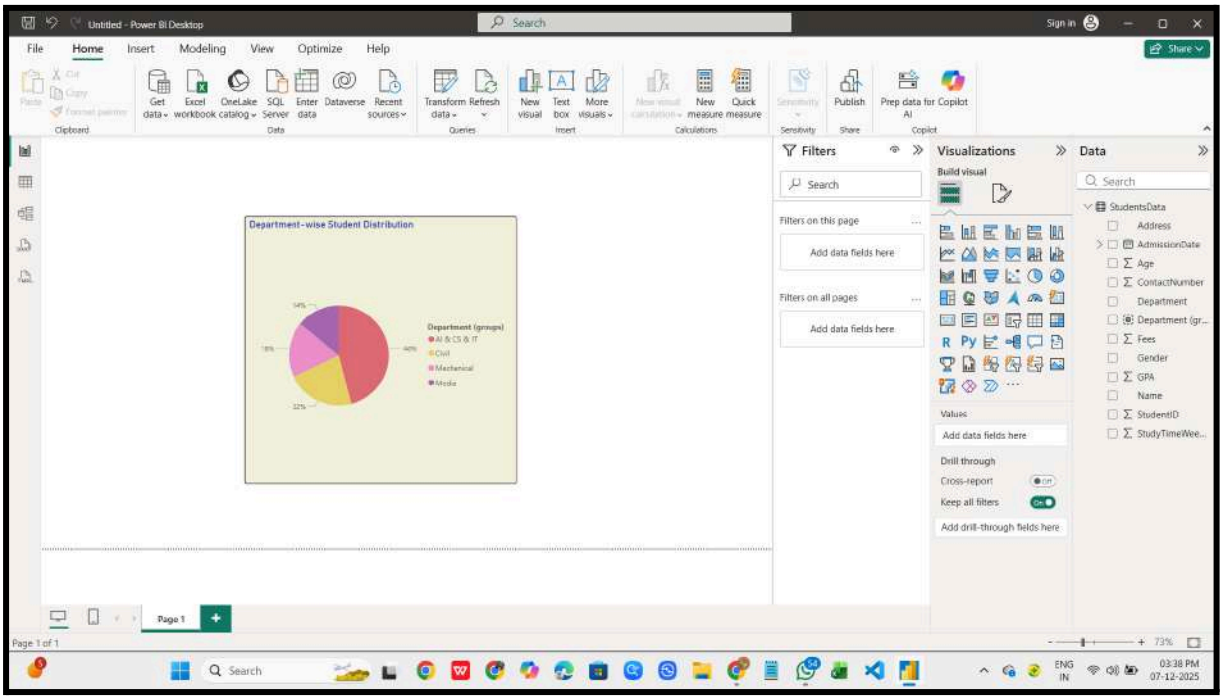



Fig. 10: Saving the File

# OUTPUT:

A fully formatted Pie Chart showing **Gender-wise distribution of students.**



# PRACTICAL NO. 3

## Aim : To create and format a Stacked Column Chart.

### Step 1: Start Power BI Desktop

- When Power BI Desktop is opened, the **Home screen** appears.

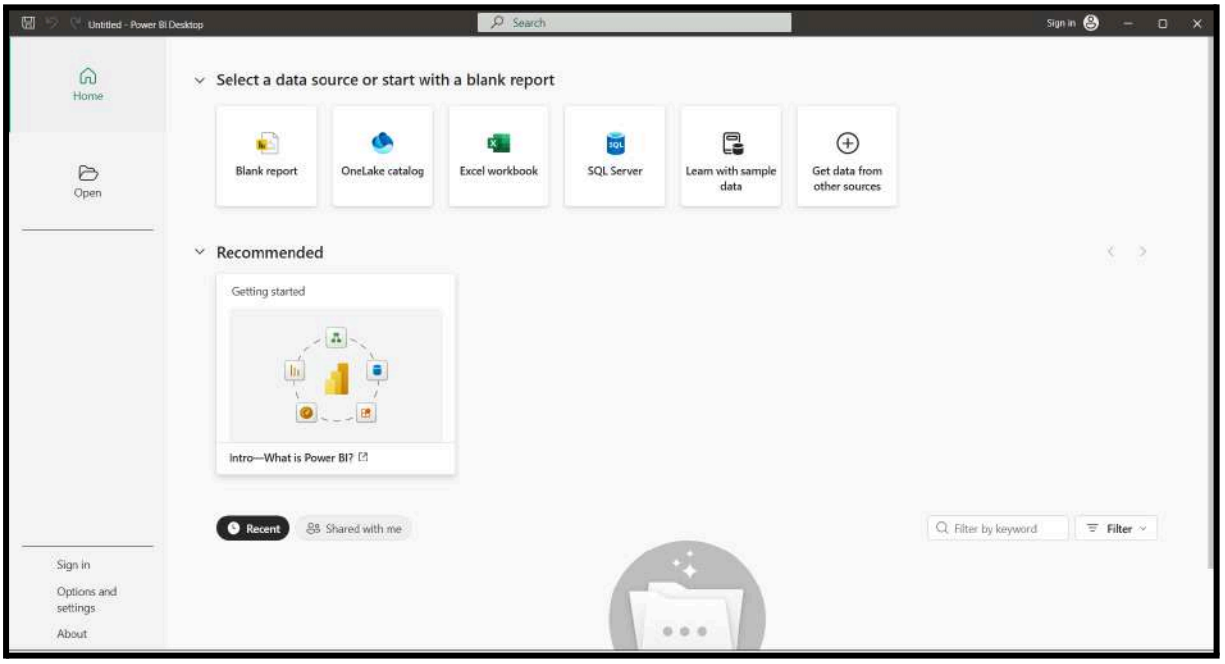



Fig. 1: Power BI Desktop Home Window

### Step 2: Import Data

- Click Excel Workbook from the Home screen.
- Select the dataset file → Click **Open**.
- A **Navigator Window** appears showing the sheets inside the workbook.

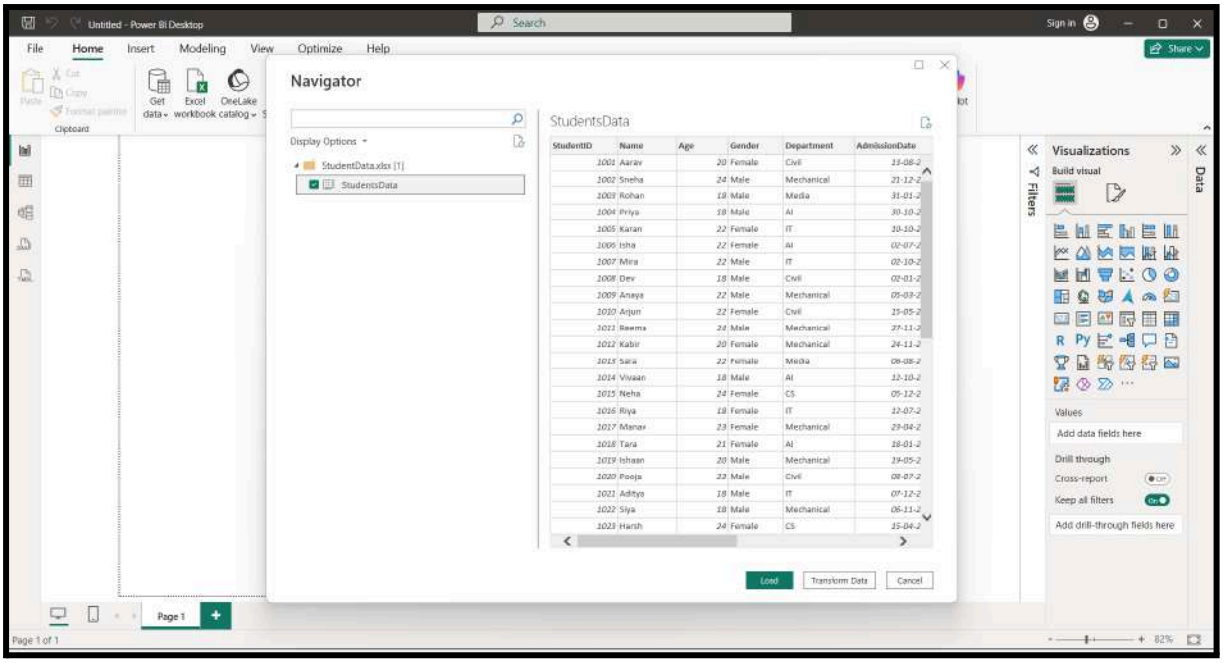



Fig. 2: Power BI Desktop Navigator Window

**Step 3: Load the Data**

- After selecting a sheet:
  - Click **Load** (if data is clean)
  - OR click **Transform Data** → Apply cleaning → Click **Close & App**
- The dataset fields now appear in the **Fields Pane** on the right side.

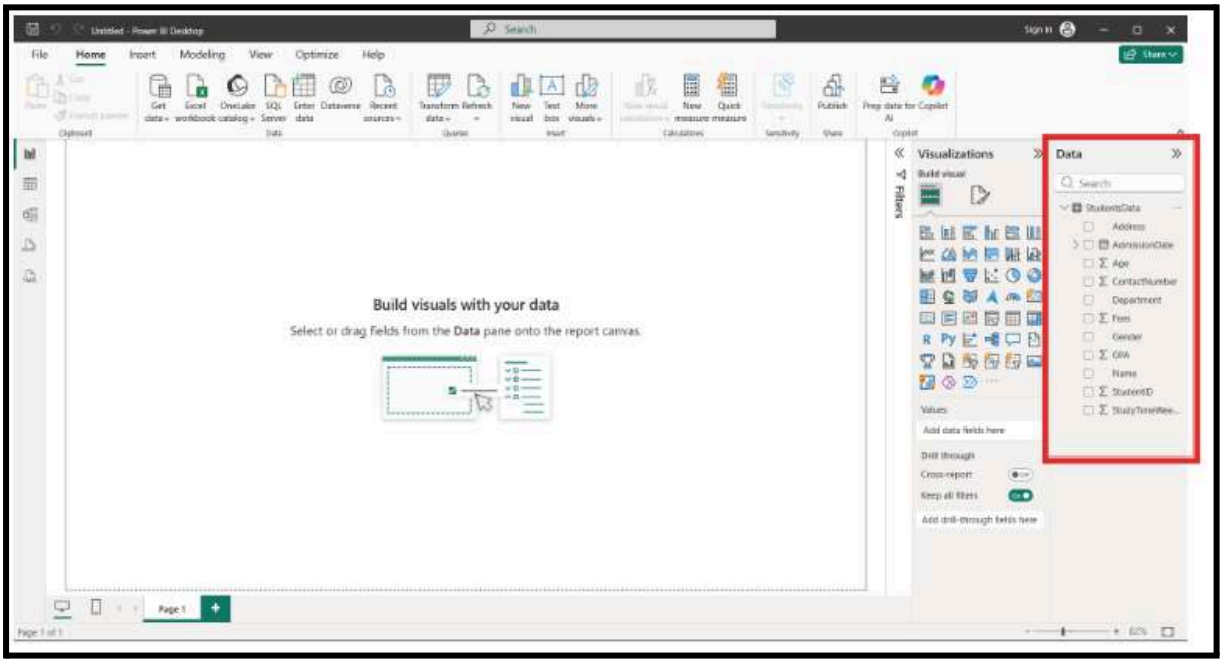
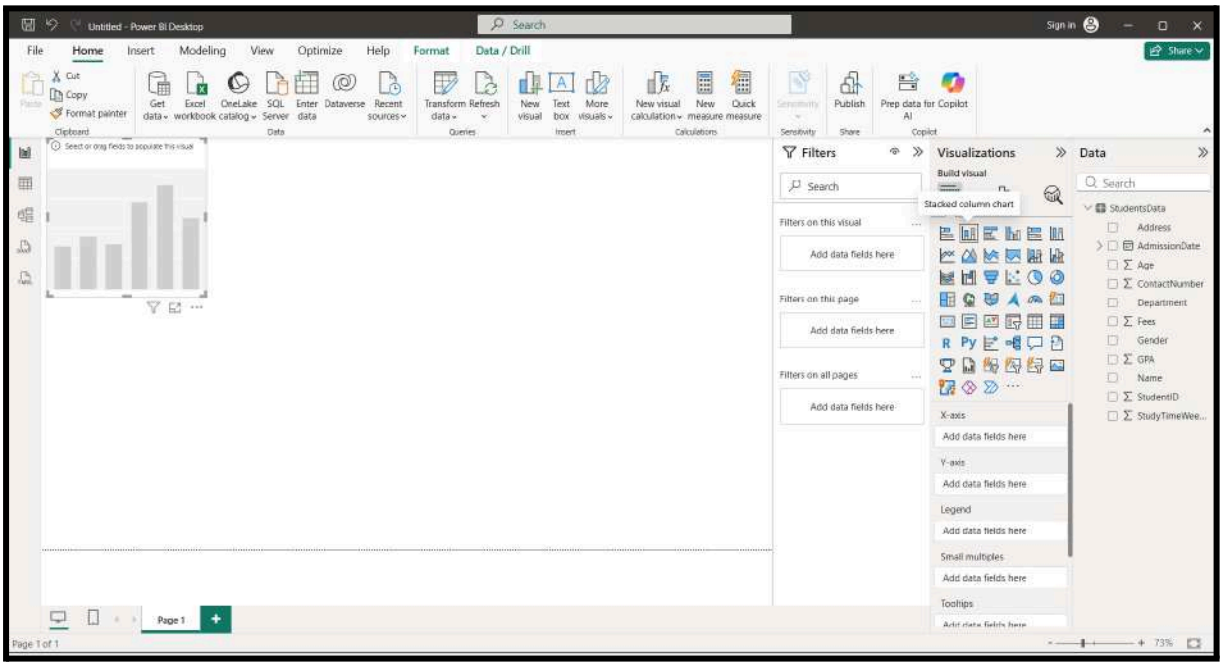



Fig. 3: Loaded Fields in Power BI

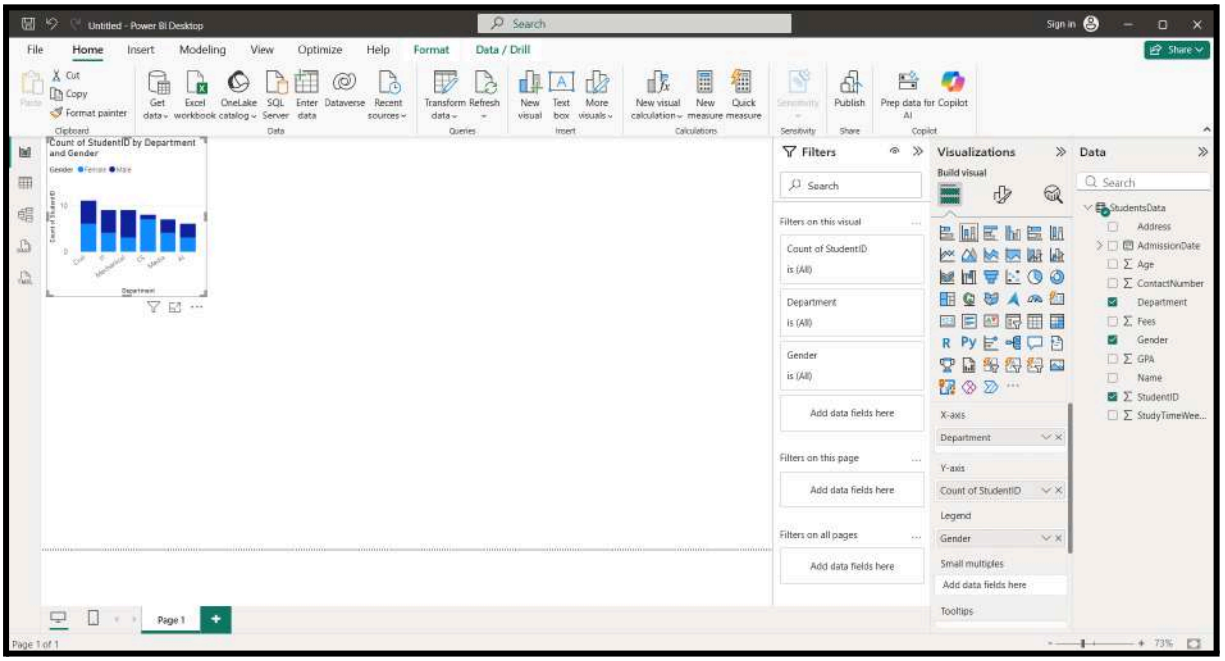
**Step 4: Insert Stacked Column Chart**

1. Go to **Visualizations Pane**
2. Click **Stacked Column Chart** icon
3. A blank chart appears on the canvas




**Step 5: Assign Fields to Stacked Column Chart**

1. Drag **Department** → Axis
2. Drag **StudentID** → Values → Click dropdown → change to **Count**
3. Drag **Gender** → Legend

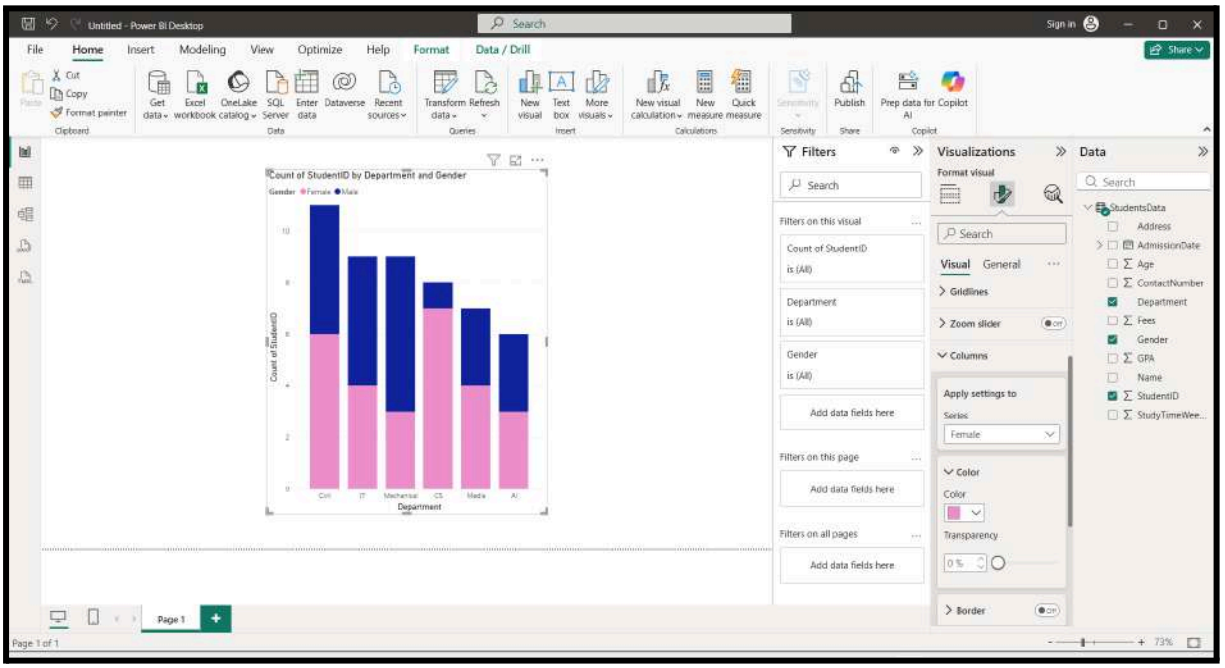


**Step 6: Format the Stacked Column Chart**

Click on the chart → Open **Format (paint roller icon)**.

**6.1 Change Bar Colors (Male/Female)**

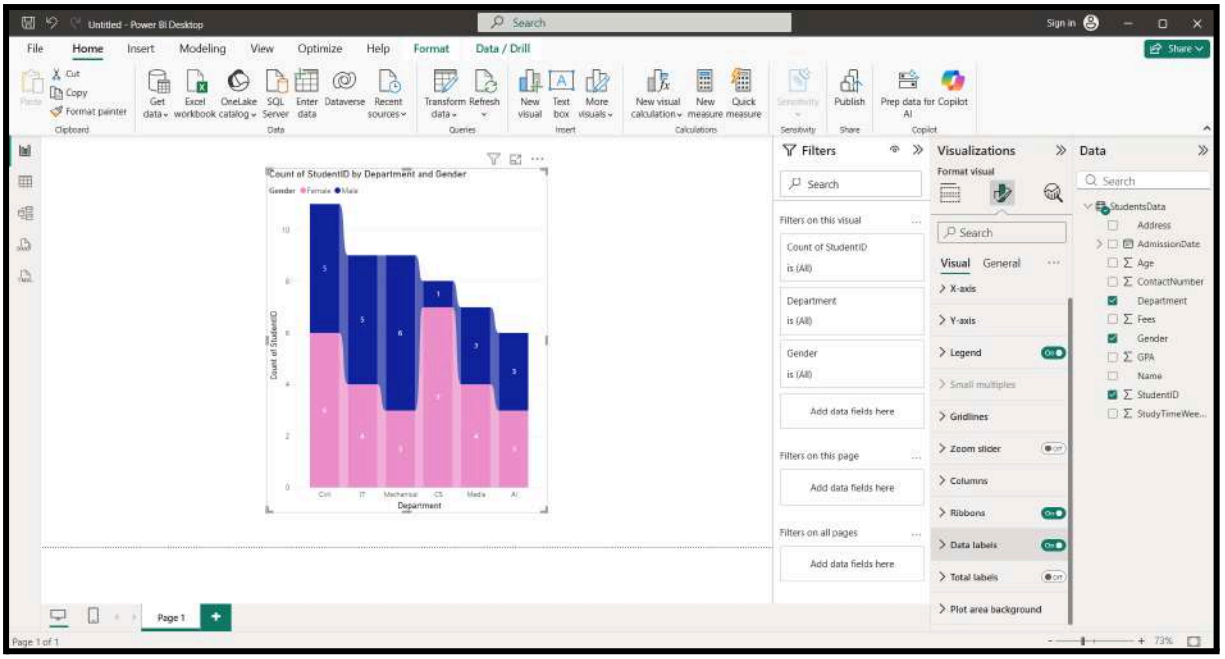
- Format → **Columns → Data Colors**
- Assign:
    - Male → Light Blue / Dark Blue
    - Female → Pink / Purple




**6.2 Add Data Labels and Ribbons**

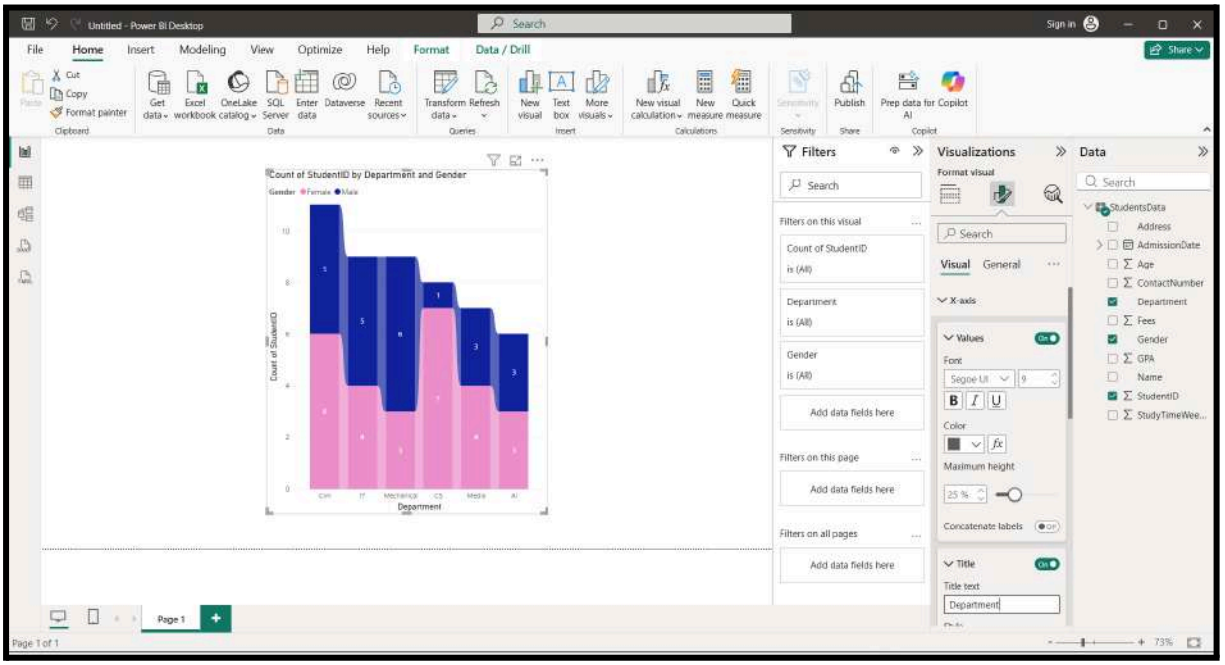
- Format → **Data Labels → On**

- Format → **Ribbons → On**



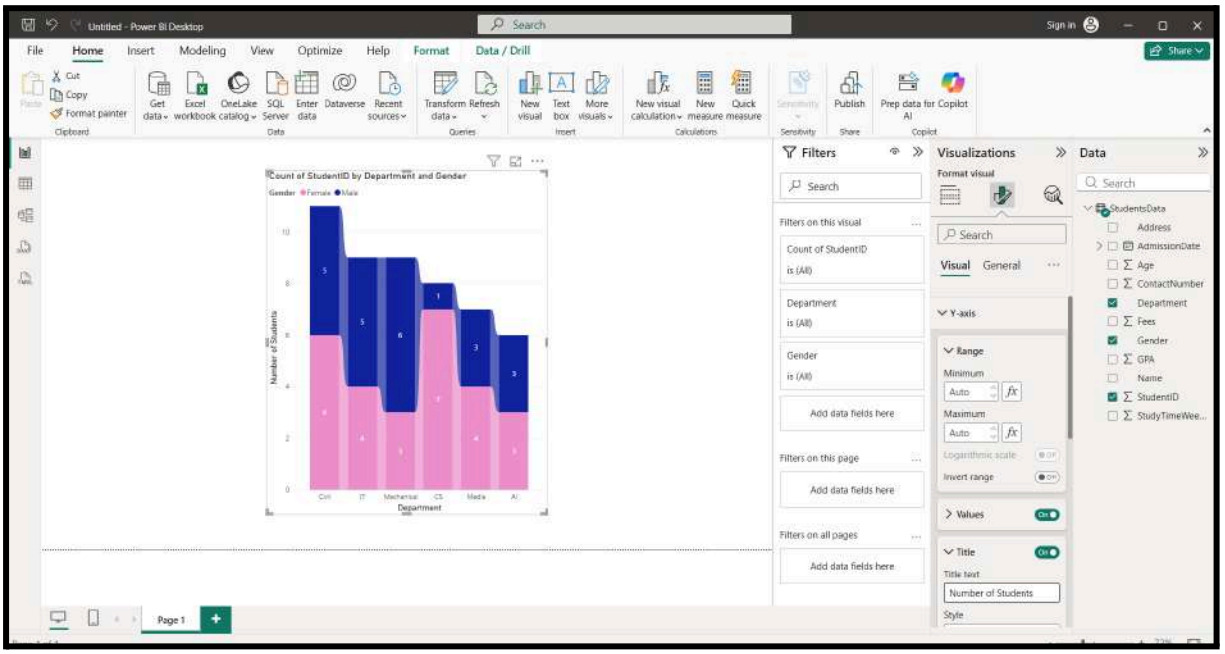

## 6.3 Format X-Axis

- Format → **X-Axis**
    - Turn **Title → On**
    - Title text: *Department*



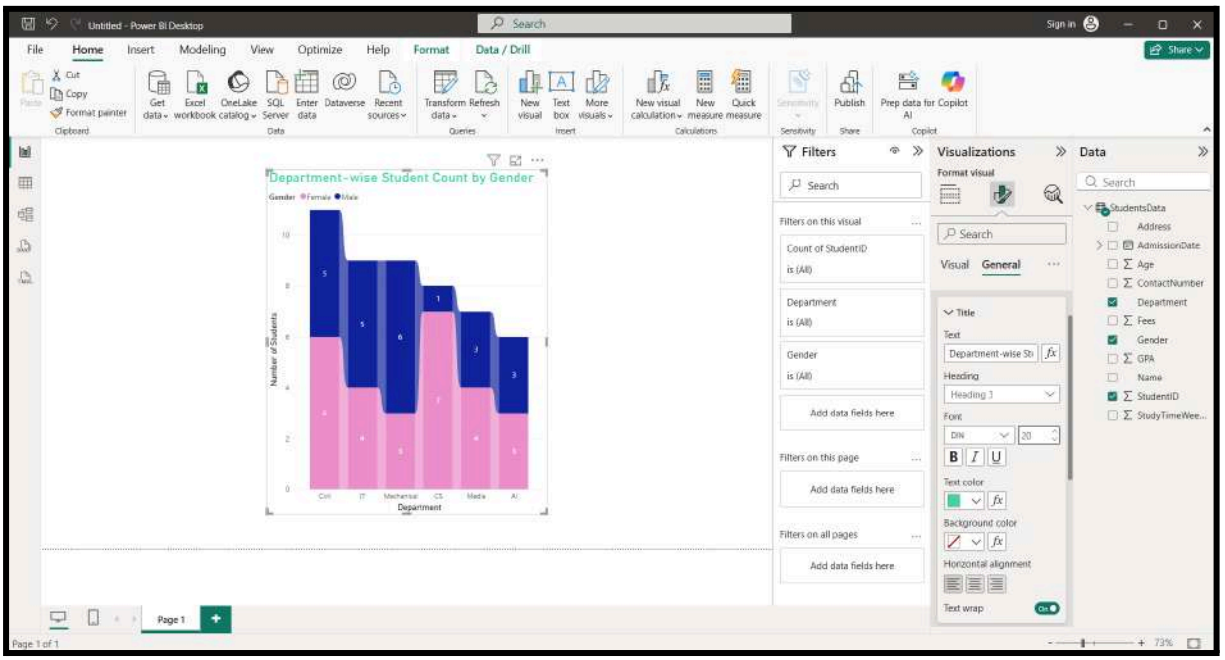

## 6.4 Format Y-Axis

- Format → **Y-Axis**
    - Turn **Title → On**
    - Title text: *Number of Students*



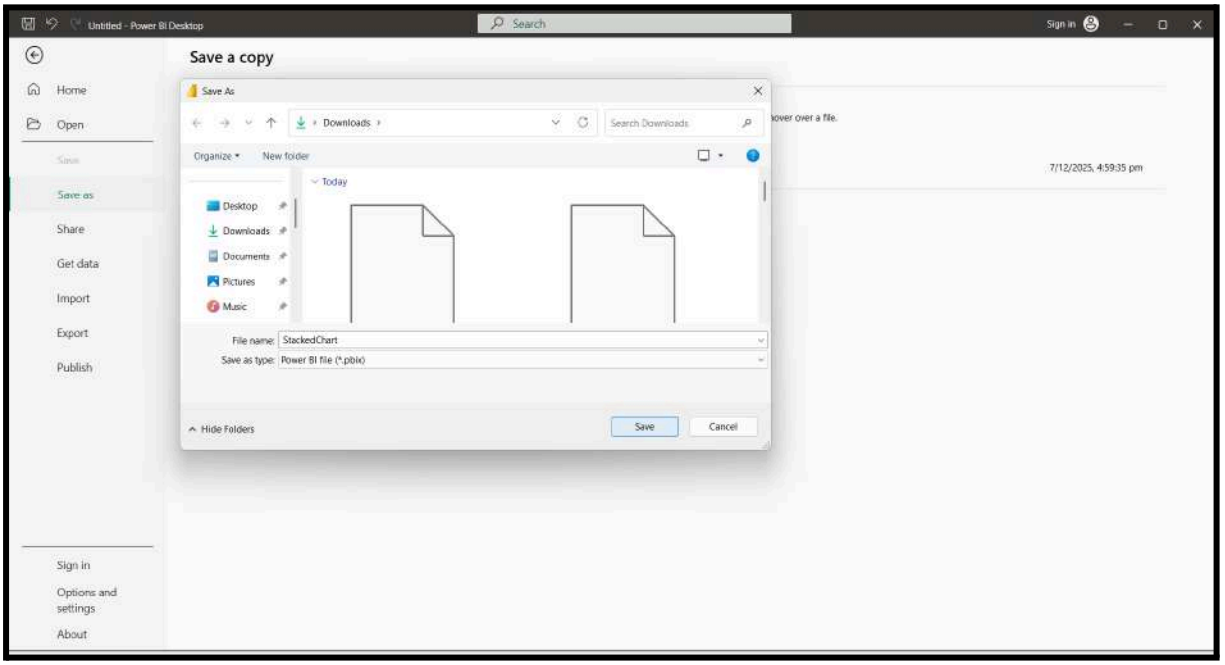

## 6.5 Format the Chart Title

- Format → **General → Title → On**
- Set Title: **"Department-wise Student Count by Gender"**



**Step 7: Save the Report**

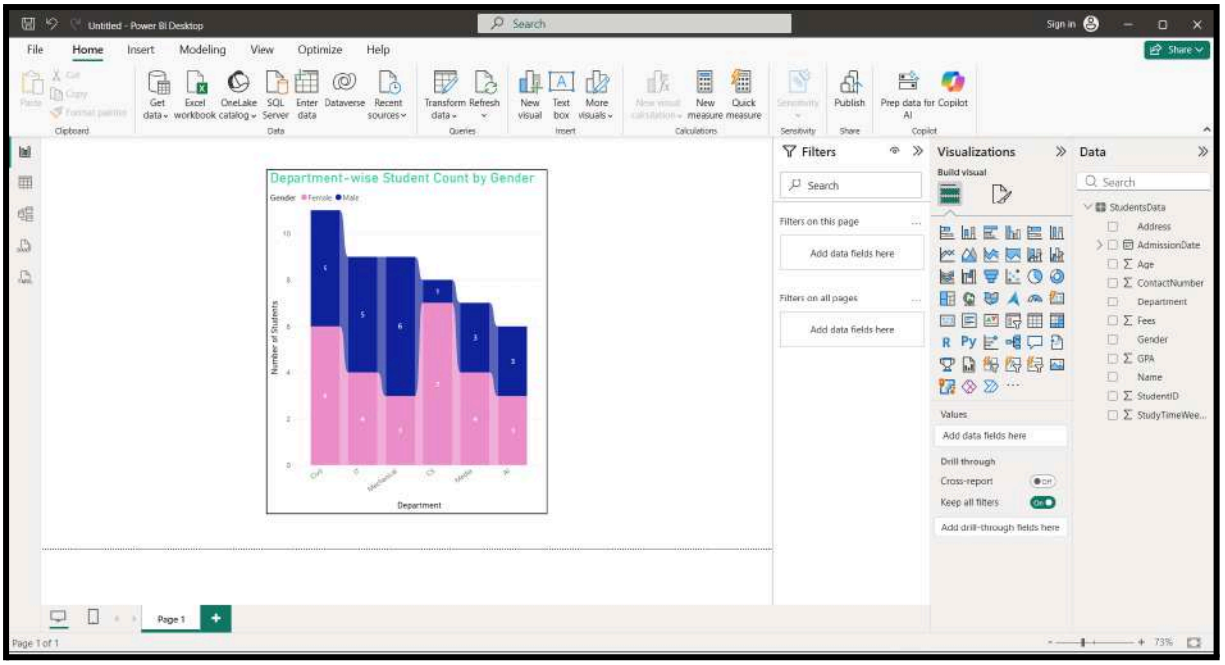
- Go to **File → Save As**
- Save the file




# OUTPUT:

A Stacked Column Chart showing:

- Each department
- Total number of students
- Gender distribution within each department


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
View
Optimize
Help
Share
Department-wise Student Count by Gender
Gender Female Male
Number of Students
Department
Filters
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
StudentsData
Address
AdmissionDate
Age
ContactNumber
Department
Fees
Gender
GPA
Name
StudentID
StudyTimeWee...
Page 1
Page 1 of 1

# PRACTICAL NO. 4

## Aim : To generate a Pie Chart inside Power BI using Python scripting and student data.

### Step 1: Check Python Version

- Open *Command Prompt* and run:

  *python --version*

### Step 2: Check PIP Version

  *pip --version*

### Step 3: Install Required Libraries

- Power BI needs these 2 libraries for Pie Charts:

  *pip install pandas*

  *pip install matplotlib*

### Step 4: Start Power BI Desktop

- When Power BI Desktop is opened, the **Home screen** appears.

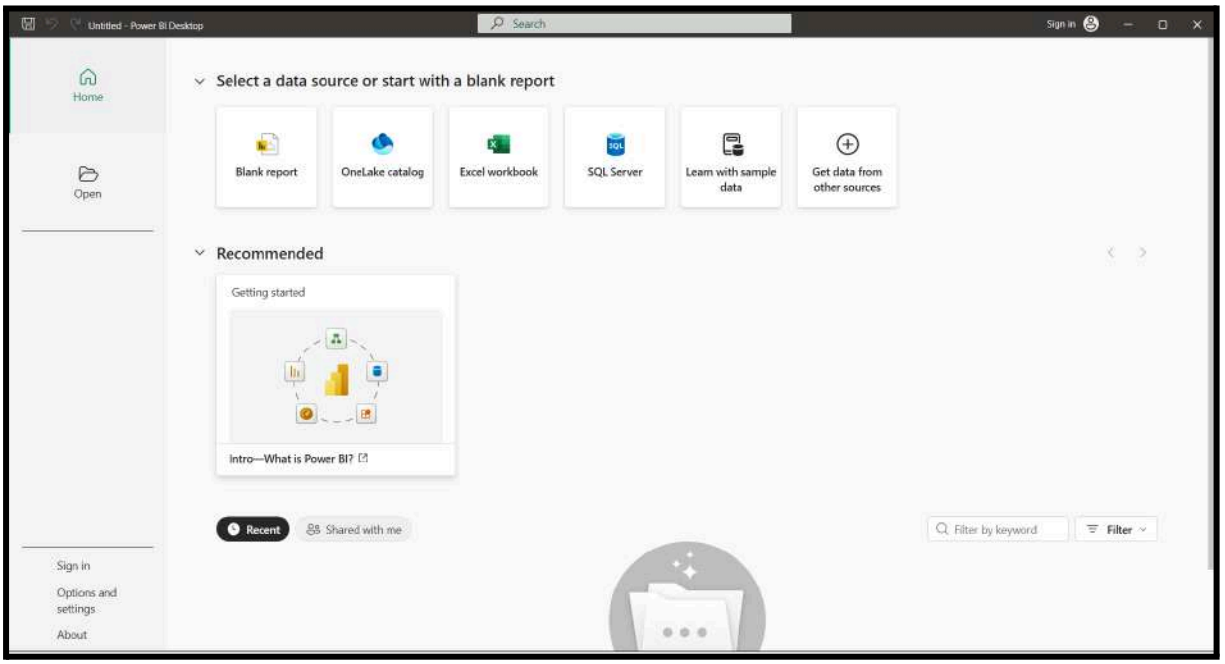



Fig. 1: Power BI Desktop Home Window

**Step 5: Import Data**

- Click Excel Workbook from the Home screen.
- Select the dataset file → Click **Open**.
- A **Navigator Window** appears showing the sheets inside the workbook.

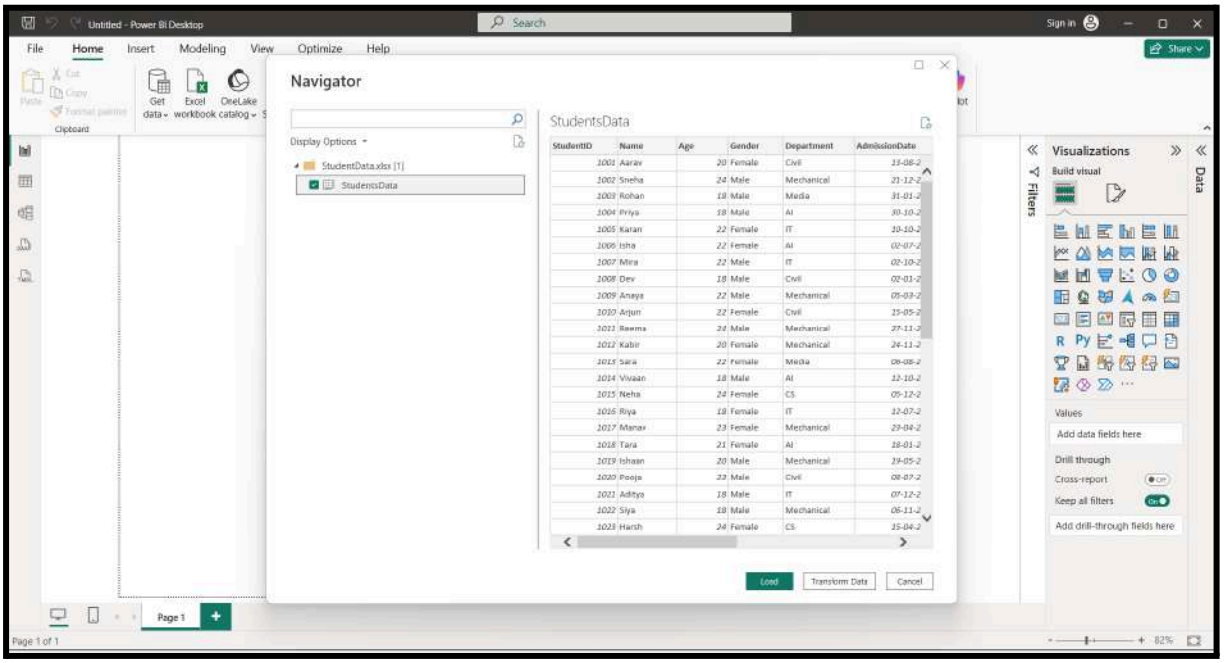



Fig. 2: Power BI Desktop Navigator Window

**Step 6: Load the Data**

- After selecting a sheet:
  - Click **Load** (if data is clean)
  - OR click **Transform Data** → Apply cleaning → Click **Close & App**
- The dataset fields now appear in the **Fields Pane** on the right side.

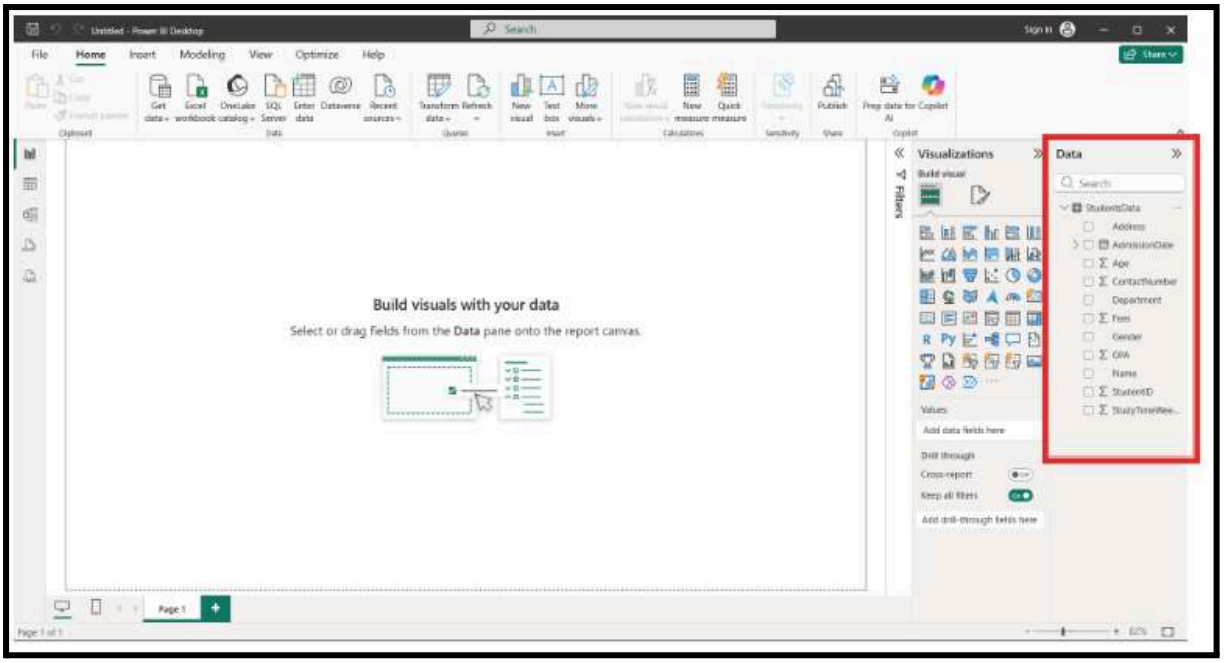
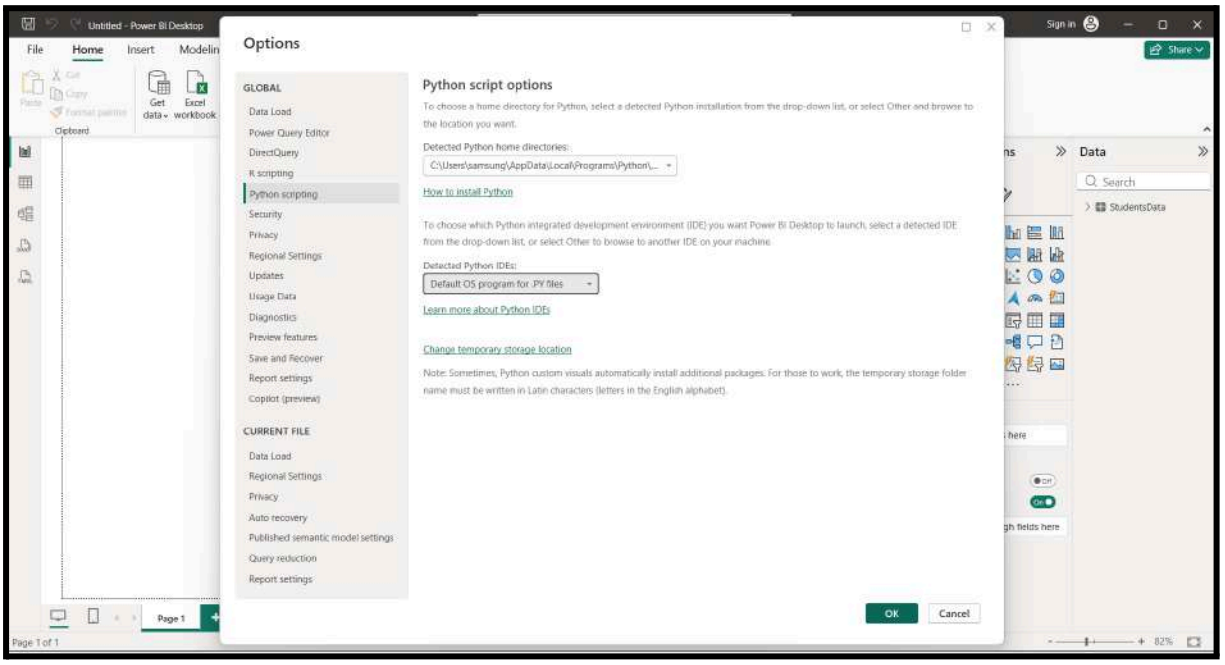



Fig. 3: Loaded Fields in Power BI

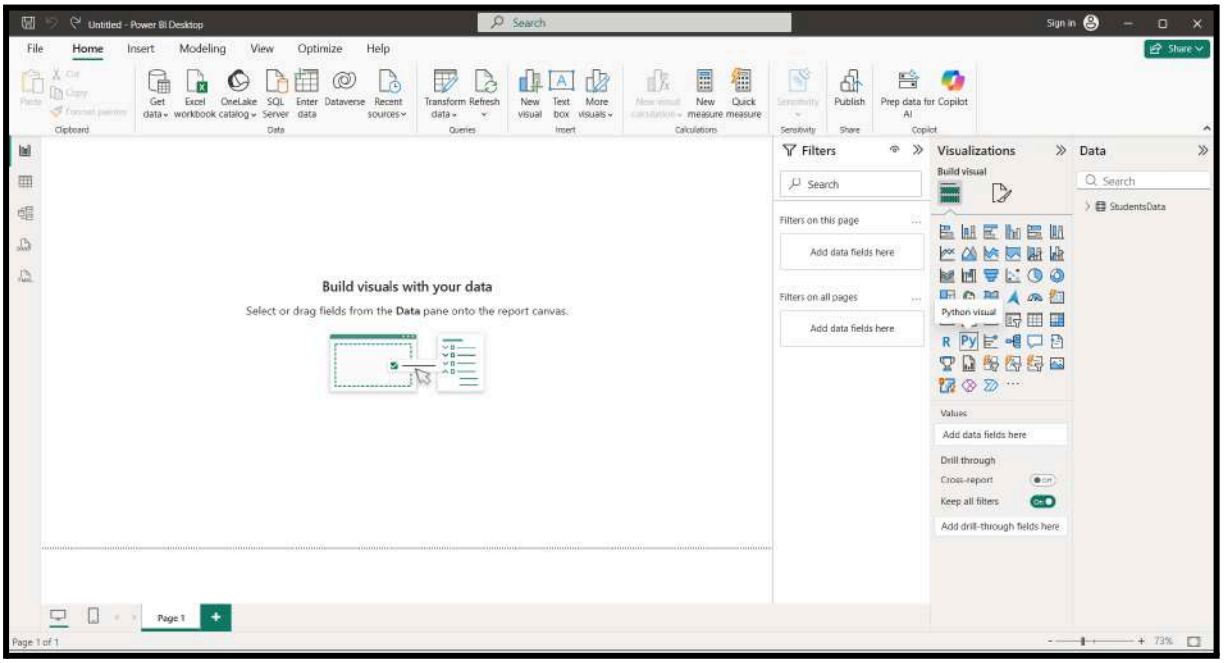
**Step 7: Enable Python Scripting in Power BI**

- Go to **File → Options and Settings → Options**
- Scroll to **Python Scripting**
- Under Python Home Directory → Select your Python installation folder
- Click **OK**

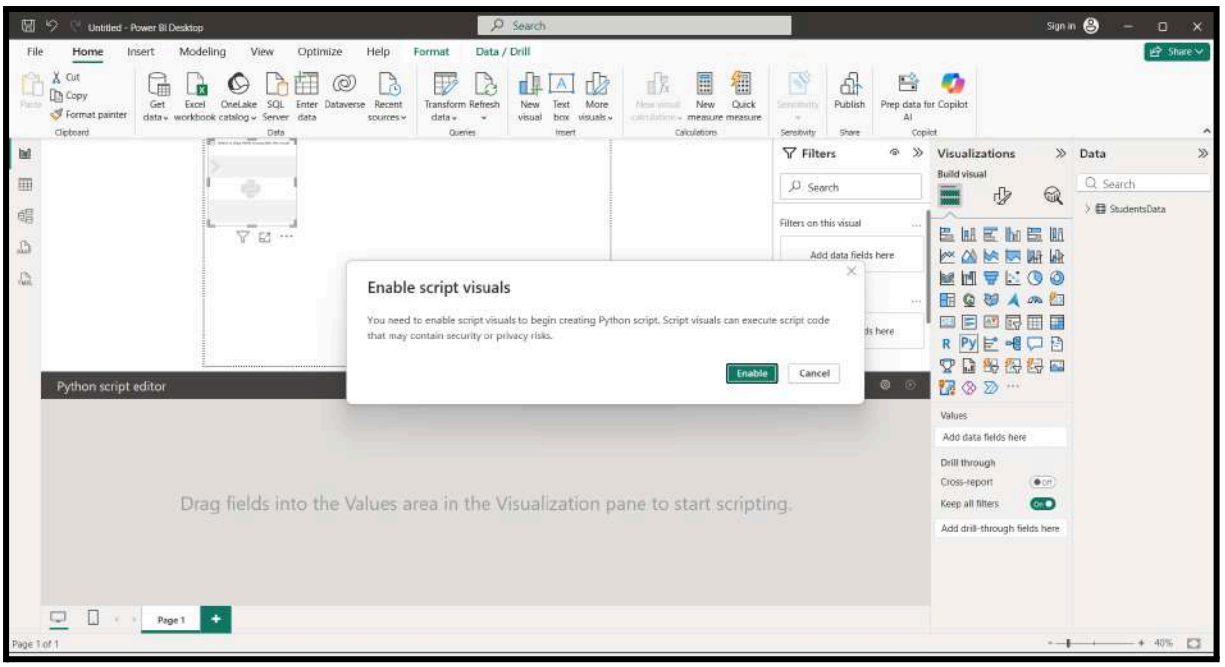



**Step 8: Add Python Visual**

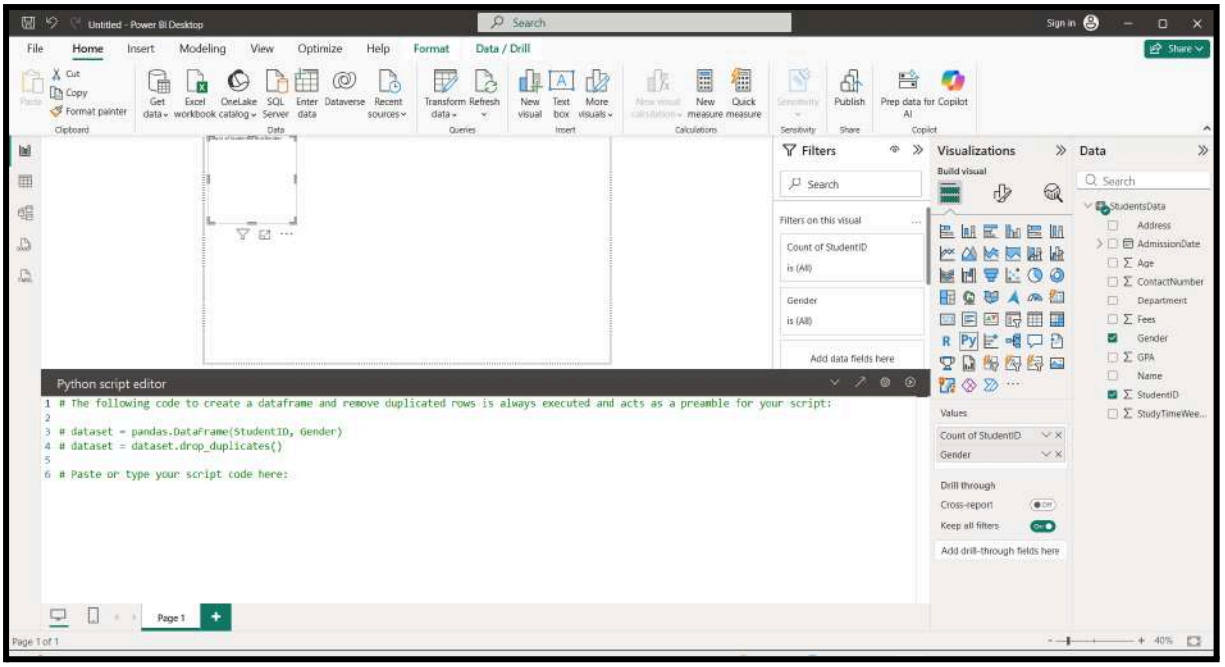
- In **Visualizations Pane**, click the **Python Visual icon**




- A blank Python script area appears on the canvas



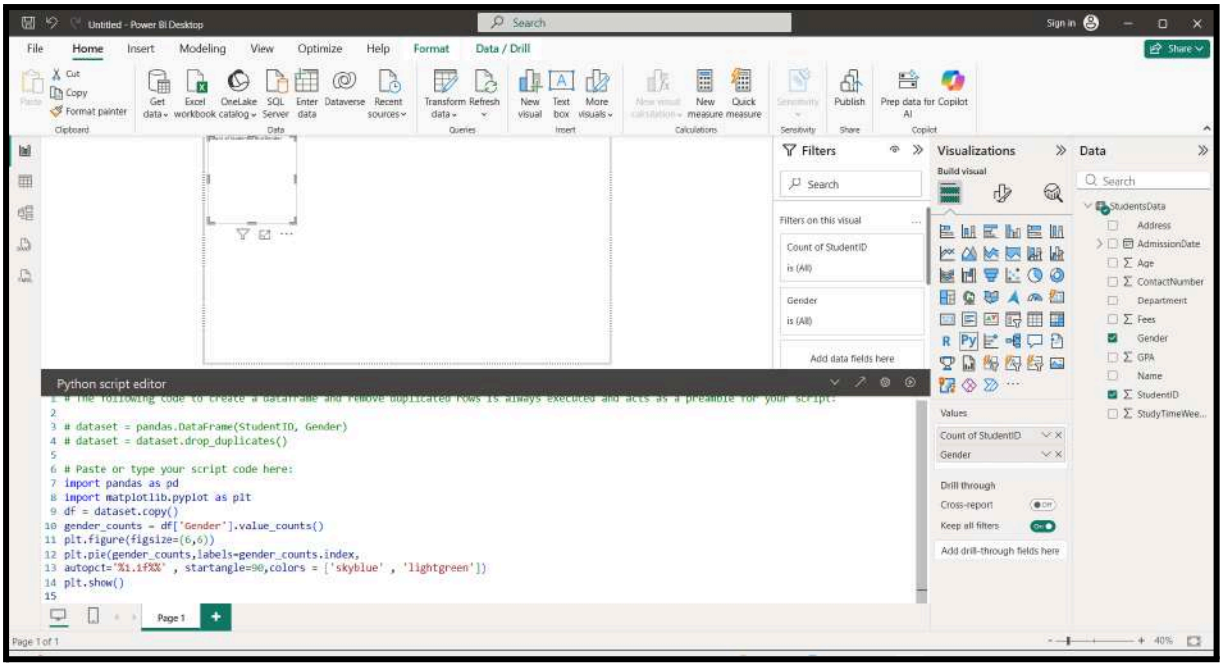

- Drag the fields you need into **Values**
- **Drag:**
  - **Gender**
  - **StudentID**



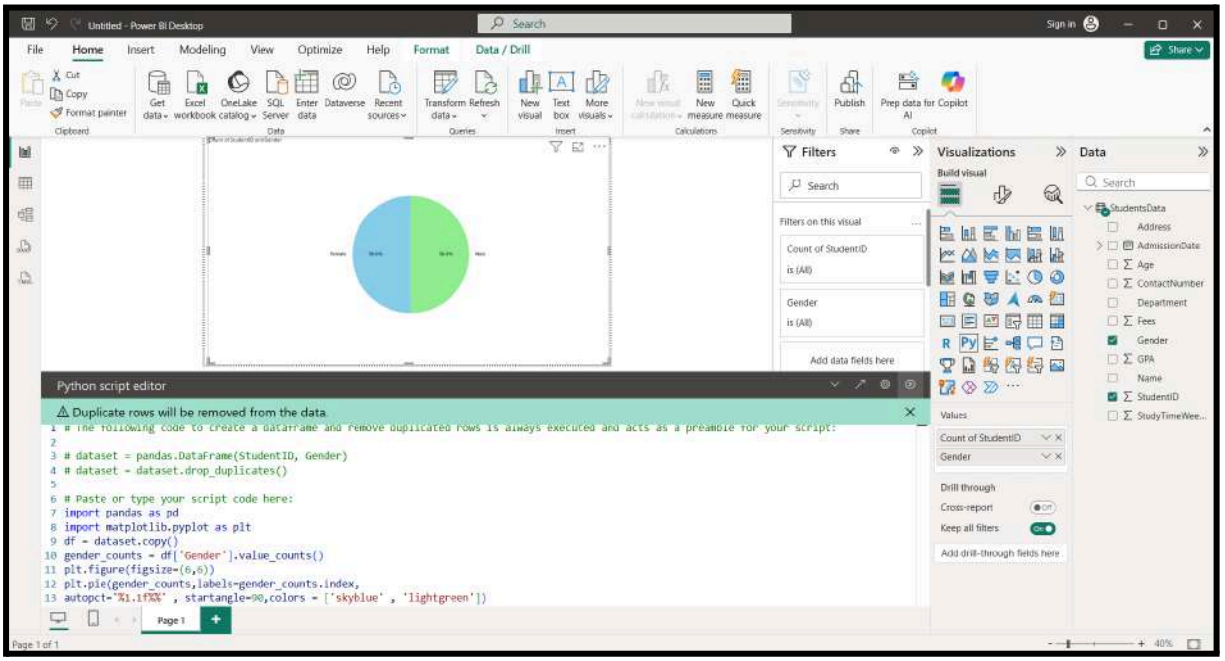

**Step 9: Write Python Script**

**Code:**

```
import pandas as pd
import matplotlib.pyplot as plt
df = dataset.copy()
gender_counts = df['Gender'].value_counts()
plt.figure(figsize=(6,6))
plt.pie(gender_counts,labels=gender_counts.index,
autopct='%1.1f%%' , startangle=90,colors = ['skyblue' , 'lightgreen'])
plt.show()
```

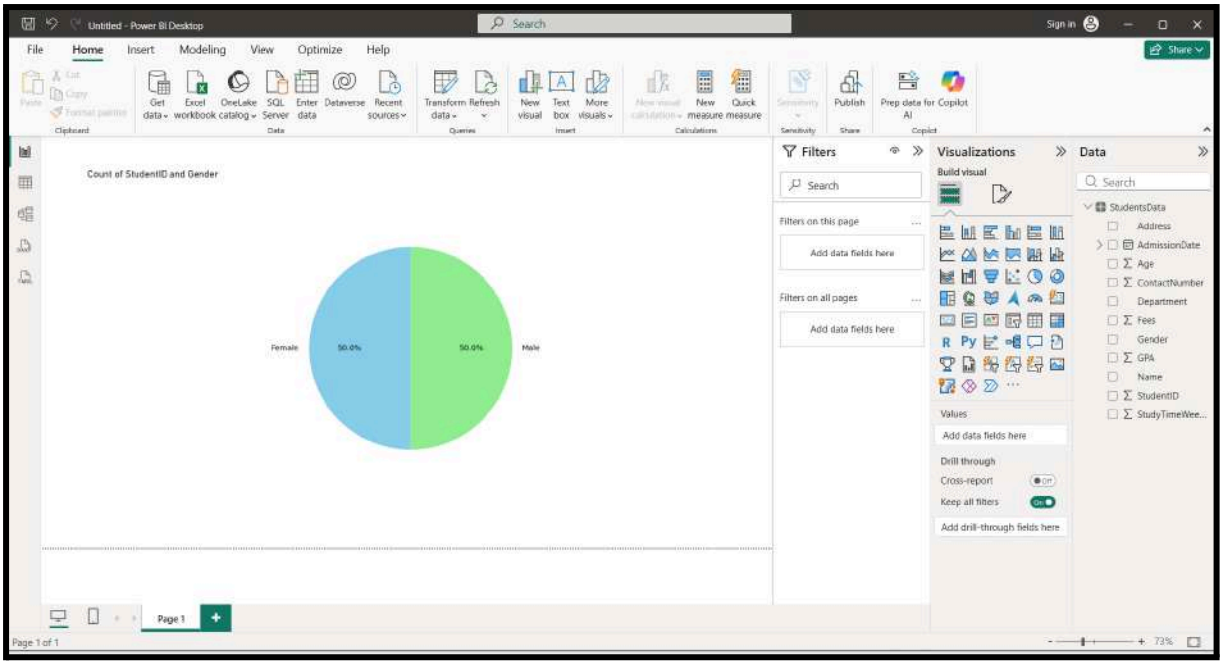



- Then press the **RUN** icon




# OUTPUT:

A Pie Chart showing **Gender-wise distribution of students** generated using Python inside Power BI.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Count of StudentID and Gender
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50.0%
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Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
Data
StudentsData
Address
AdmissionDate
Age
ContactNumber
Department
Fees
Gender
GPA
Name
StudentID
StudyTimeWee...
Page 1
Page 1 of 1
73%

# PRACTICAL NO. 5

## Aim : To create a scatter plot inside Power BI using Python scripting.

### Step 1: Check Python Version

- Open *Command Prompt* and run:

  *python --version*

### Step 2: Check PIP Version

  *pip --version*

### Step 3: Install Required Libraries

- Power BI needs these 2 libraries for Pie Charts:

  *pip install pandas*

  *pip install matplotlib*

### Step 4: Start Power BI Desktop

- When Power BI Desktop is opened, the **Home screen** appears.

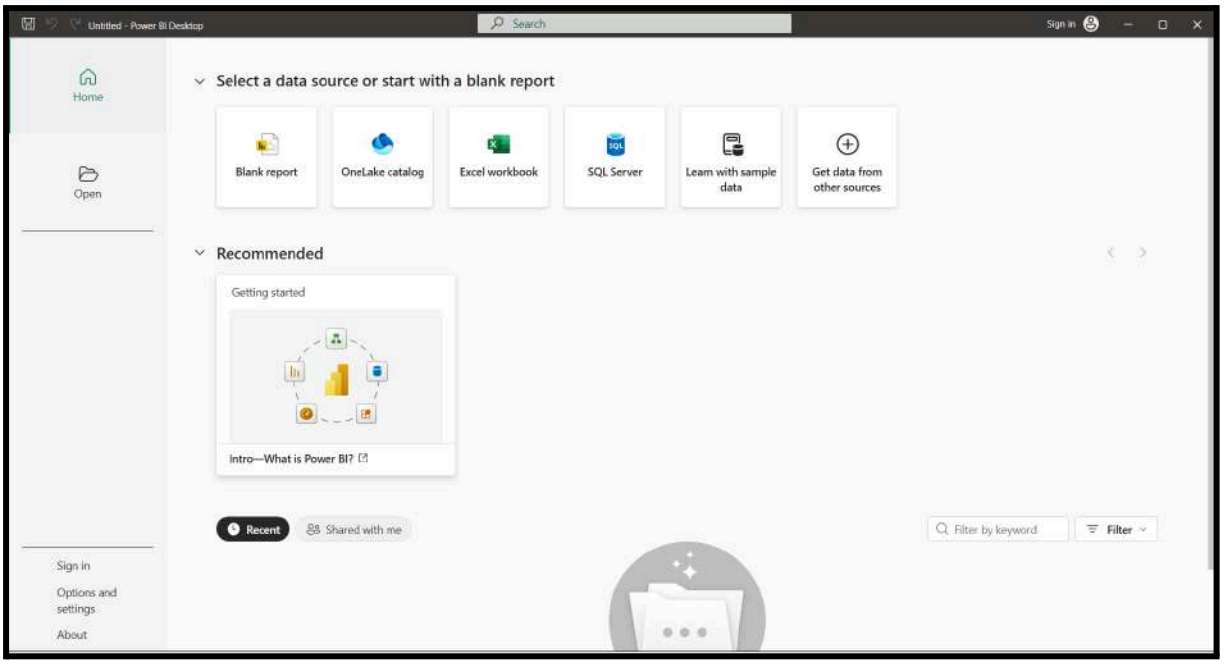



Fig. 1: Power BI Desktop Home Window

**Step 5: Import Data**

- Click Excel Workbook from the Home screen.
- Select the dataset file → Click **Open**.
- A **Navigator Window** appears showing the sheets inside the workbook.

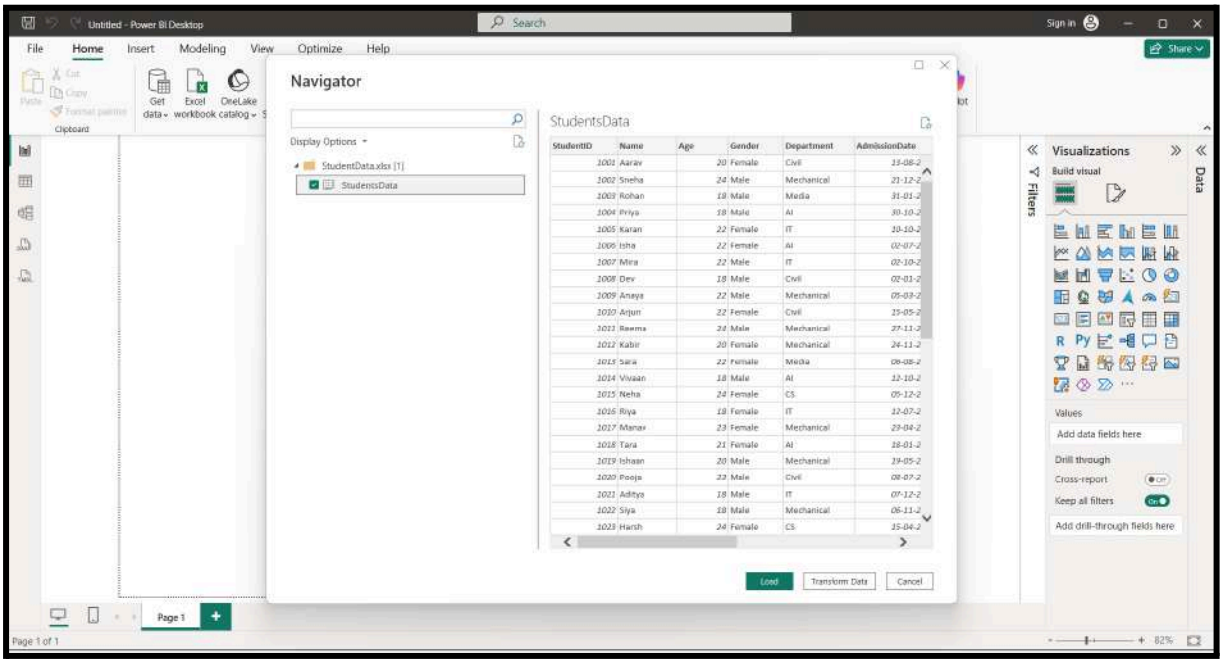



Fig. 2: Power BI Desktop Navigator Window

**Step 6: Load the Data**

- After selecting a sheet:
  - Click **Load** (if data is clean)
  - OR click **Transform Data** → Apply cleaning → Click **Close & App**
- The dataset fields now appear in the **Fields Pane** on the right side.

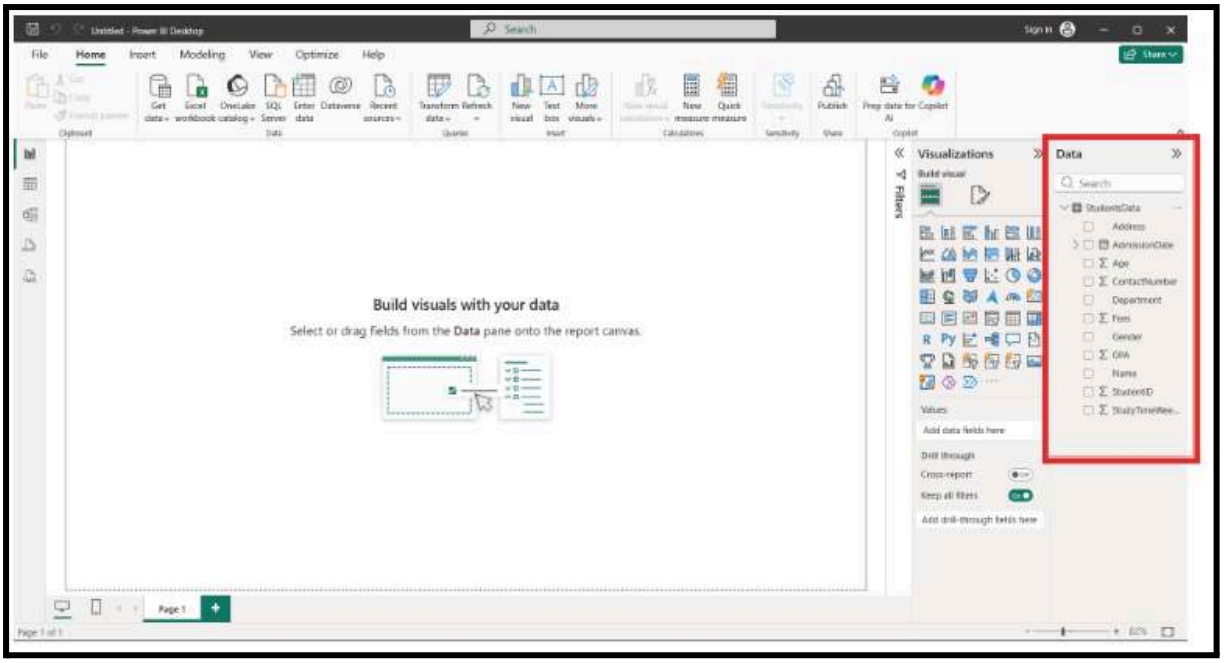
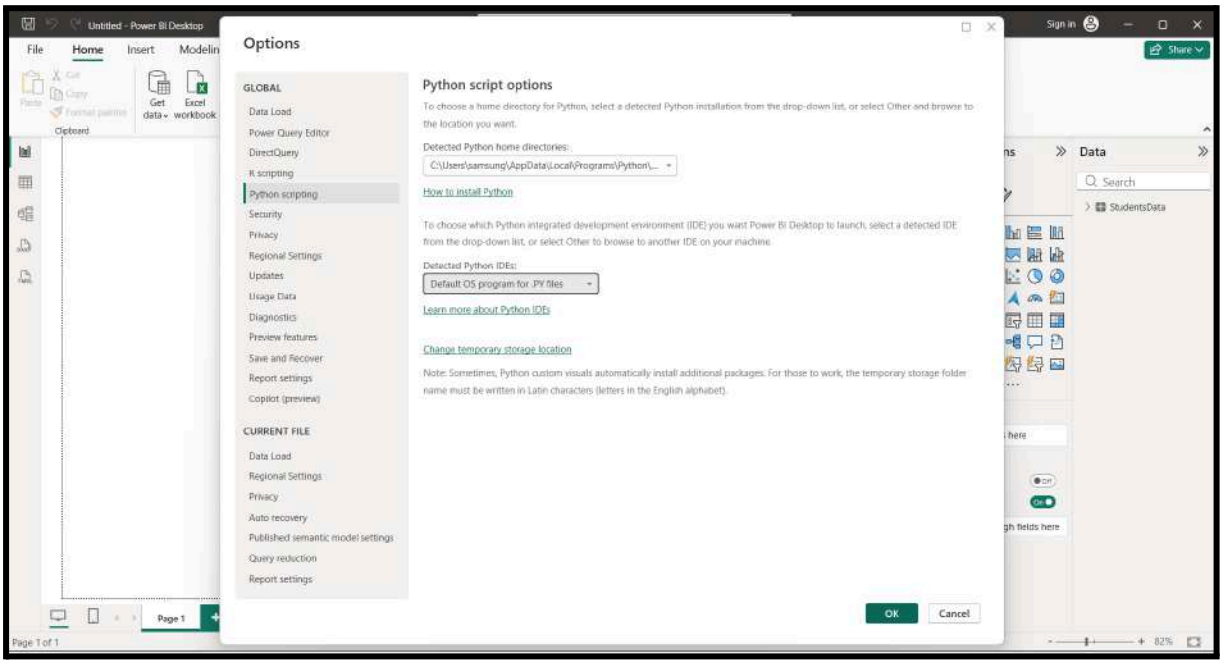



Fig. 3: Loaded Fields in Power BI

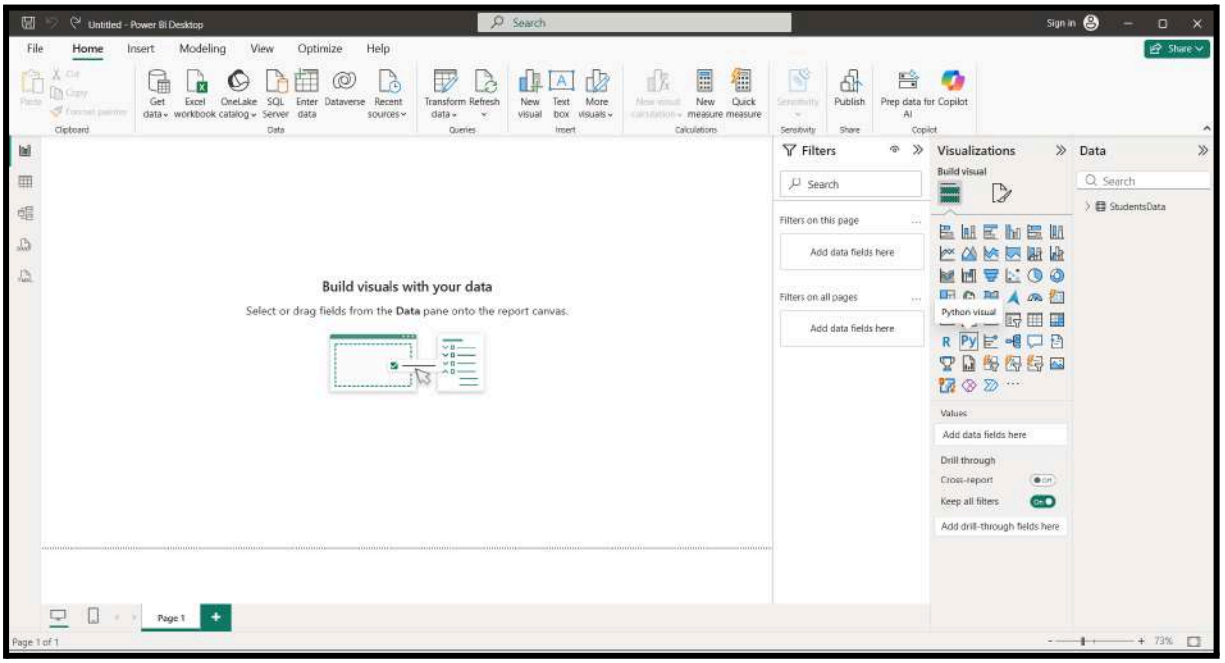
**Step 7: Enable Python Scripting in Power BI**

- Go to **File → Options and Settings → Options**
- Scroll to **Python Scripting**
- Under Python Home Directory → Select your Python installation folder
- Click **OK**

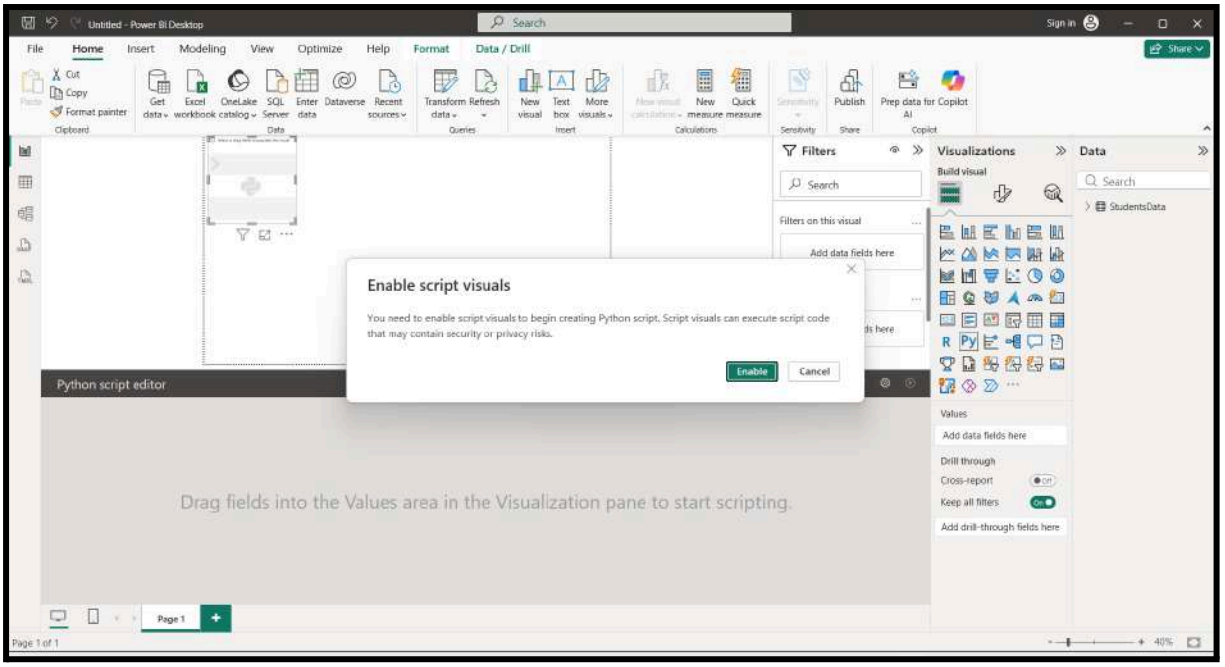



**Step 8: Add Python Visual**

- In **Visualizations Pane**, click the **Python Visual icon**

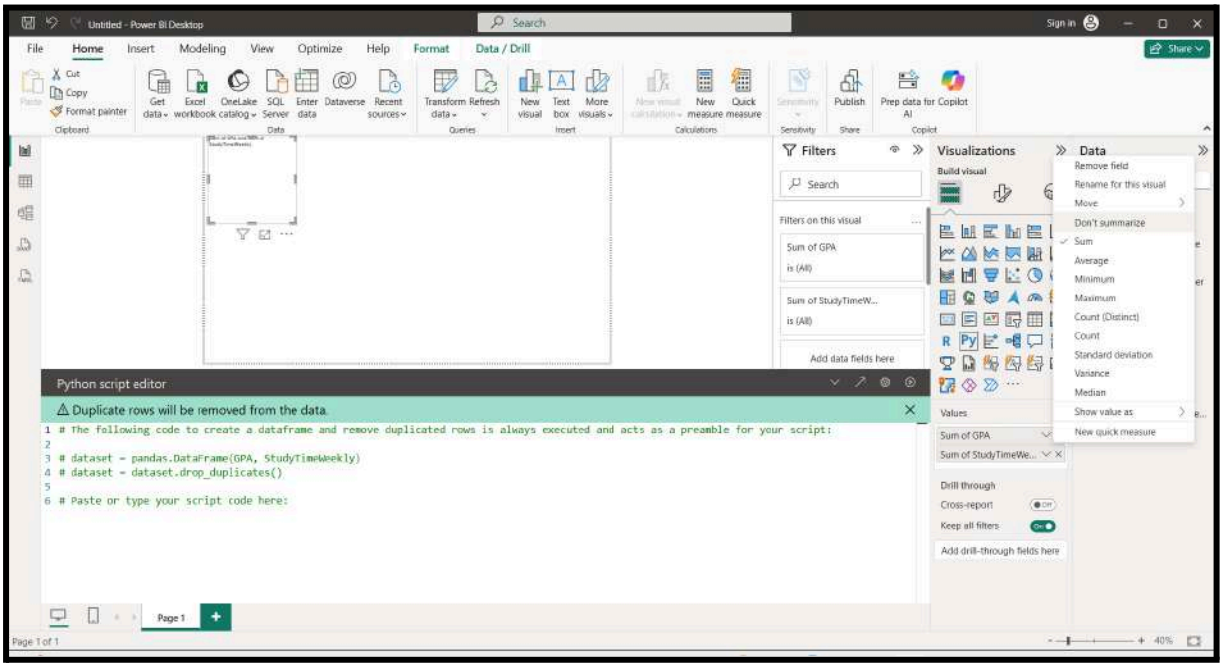



- A blank Python script area appears on the canvas




- Drag the fields you need into **Values**
- **Drag:**
  - **StudyTimeWeekly**
    - **Click the dropdown**
    - **Choose Don't summarize**

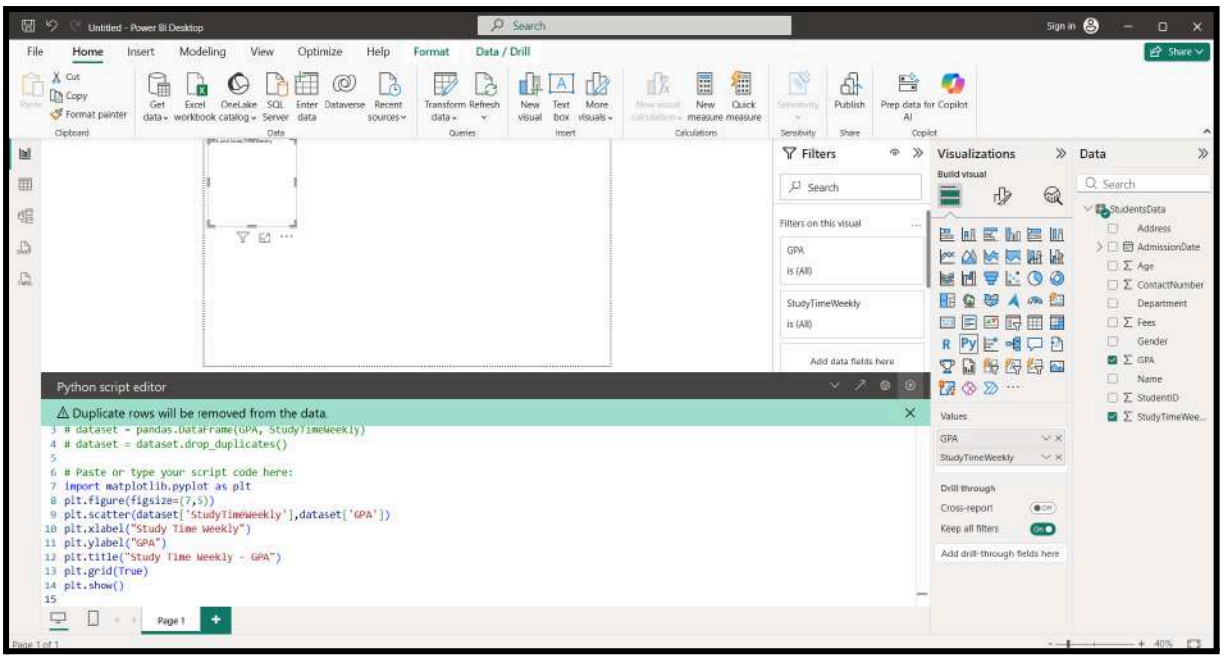
- **GPA**
    - **Click the dropdown**
    - **Choose Don't summarize**



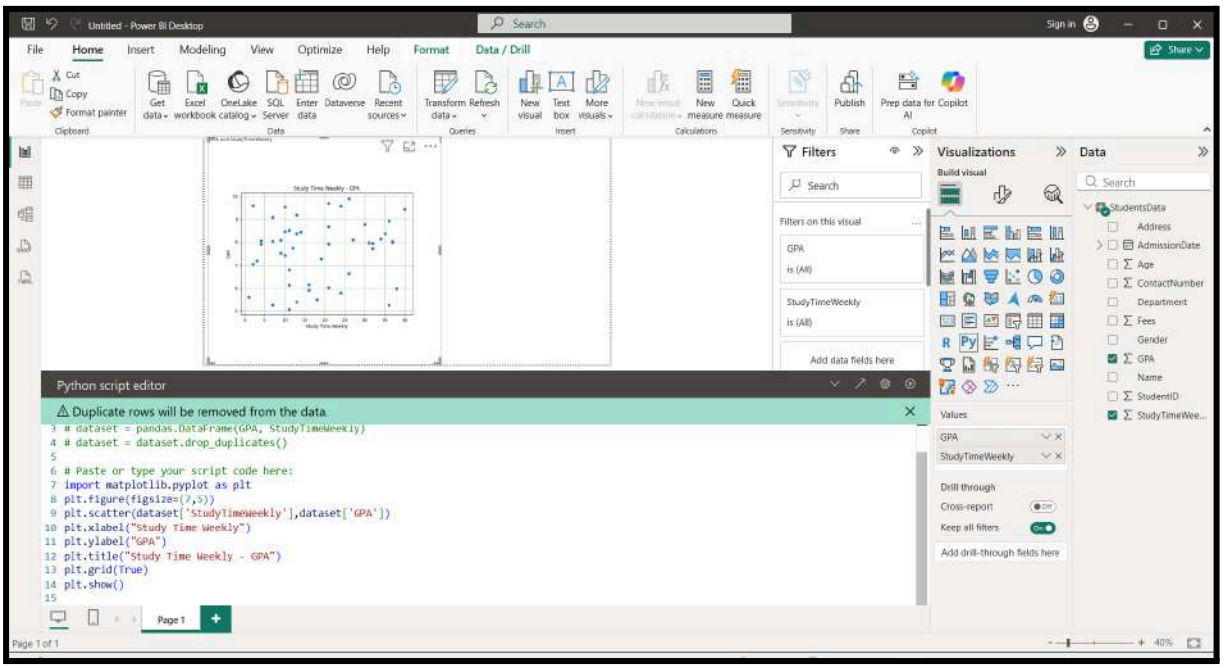

**Step 9: Write Python Script**

**Code:**

```
import matplotlib.pyplot as plt
plt.figure(figsize=(7,5))
plt.scatter(dataset['StudyTimeWeekly'],dataset['GPA'])
plt.xlabel("Study Time Weekly")
plt.ylabel("GPA")
plt.title("Study Time Weekly - GPA")
plt.grid(True)
plt.show()
```

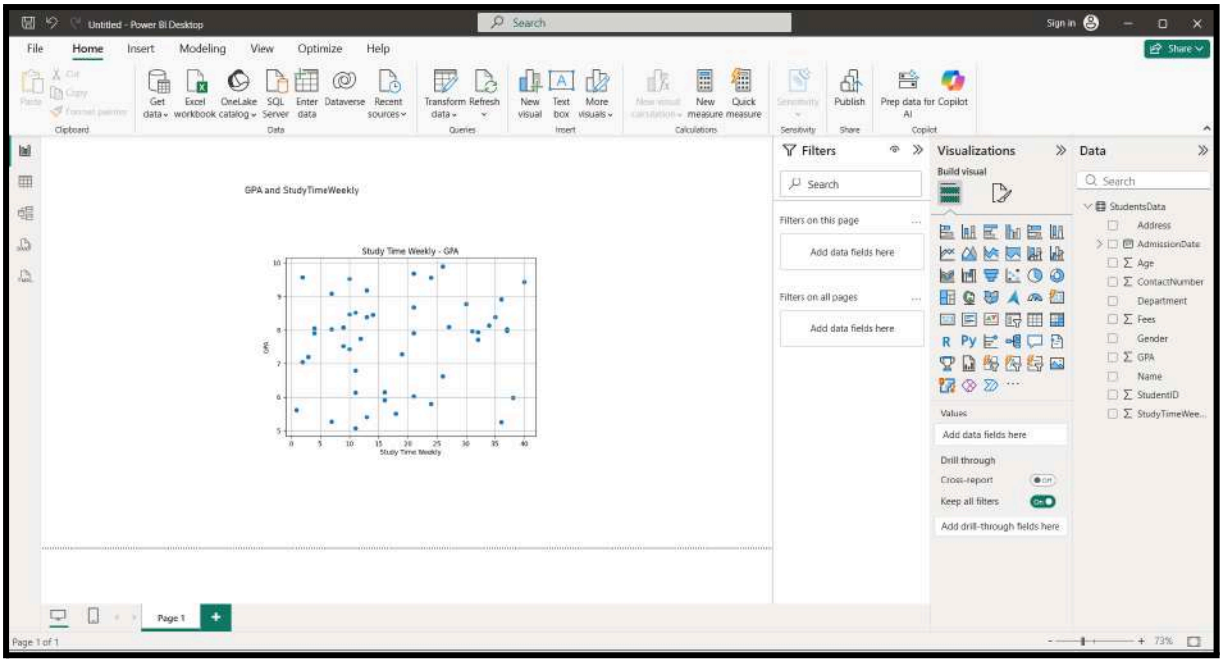



- Then press the **RUN** icon




# OUTPUT:

A Scatter Chart showing the relationship between **Study Time Weekly** and **GPA**, generated using Python inside Power BI.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Search
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File
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View
Optimize
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Clipboard
Data
Queries
Insert
Calculations
Sensitivity
Share
Copilot
Publish
Prep data for AI
GPA and StudyTimeWeekly
Study Time Weekly - GPA
Study Time Weekly
Filters
Filters on this page
Add data fields here
Filters on all pages
Add data fields here
Visualizations
Build visual
Values
Add data fields here
Drill through
Cross-report
Keep all filters
Add drill-through fields here
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StudentsData
Address
AdmissionDate
Age
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Department
Fees
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Page 1
Page 1 of 1
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